Advances in artificial spin ice

Sandra H. Skjærvø[1,2]*, Christopher H. Marrows[3], Robert L. Stamps[4] and Laura J. Heyderman[1,2]*

Abstract | Artificial spin ices consist of nanomagnets arranged on the sites of various periodic and aperiodic lattices. They have enabled the experimental investigation of a variety of fascinating phenomena such as frustration, emergent magnetic monopoles and phase transitions that have previously been the domain of bulk spin crystals and theory, as we discuss in this Review. Artificial spin ices also show promise as reprogrammable magnonic crystals and, with this in mind, we give an overview of the measurements of fast dynamics in these magnetic metamaterials. We survey the variety of geometries that have been implemented, in terms of both the form of the nanomagnets and the lattices on which they are placed, including quasicrystalline systems and artificial spin systems in 3D. Different magnetic materials can also be incorporated to modify anisotropies and blocking temperatures, for example. With this large variety of systems, the way is open to discover new phenomena, and we complete this Review with possible directions for the future.

[1]Laboratory for Mesoscopic Systems, Department of Materials, ETH Zurich, Zurich, Switzerland.

[2]Laboratory for Multiscale Materials Experiments, Paul Scherrer Institute, Villigen, Switzerland.

[3]School of Physics and Astronomy, University of Leeds, Leeds, UK.

[4]Department of Physics and Astronomy, University of Manitoba, Winnipeg, Canada.

*e-mail: sandra.skjaervoe@psi.ch; laura.heyderman@psi.ch

https://doi.org/10.1038/s42254-019-0118-3

Artificial spin ices are metamaterials in the sense that they are engineered systems that exhibit a variety of properties, such as collective dynamics, that are not inherently present in their building blocks. The smallest building blocks in artificial spin ices are single-domain magnets with submicrometre dimensions (nanomagnets), which are arranged in specific designs and are coupled by their dipolar magnetic fields. In some artificial spin ices, the nanomagnets are physically connected, so that they are also coupled through the exchange interaction. The nanomagnets are arranged in such a way that their moments (often called macrospins) are frustrated, which means that not all interactions between the nanomagnets can be satisfied simultaneously. The mesoscopic size of the nanomagnets makes it possible to directly observe the moment configurations with magnetic microscopy techniques.

Artificial spin ices were originally developed[1] to mimic the behaviour of spins in their crystal counterparts, such as the rare-earth titanate pyrochlores[2], which have spins positioned on the corners of tetrahedra. Correspondingly, in the first studies, the nanomagnets were placed on the sites of the square or kagome lattice (FIG. 1). An early focus of interest was magnetic-field-driven phenomena, with observations reported of magnetization reversal and the use of an alternating magnetic field to perform an effective thermal anneal. Later, it became possible to create thermodynamic systems in which the energy barrier to switching was small enough to allow the moments of the nanomagnets to switch spontaneously, but in a collective manner mediated by the dipolar interaction between the nanomagnets.

In this Review, we focus on what has happened in the field of artificial spin ice since two major reviews in 2013 (REFS[3,4]), which provide a useful foundation for a reader unfamiliar with the field. In particular, in frustrated arrangements of nanomagnets, there has been a wide variety of interesting phenomena discovered, including emergent magnetic monopoles[5,6], vertex-based frustration[7,8], chiral dynamics[9] and phase transitions[10–12]. The existence of emergent phenomena, which arise from the collective behaviour of the nanomagnets, underscores why artificial spin ices can be considered as metamaterials. Artificial spin systems are also important for high-frequency dynamics because of their potential as programmable magnonic crystals[13]. Underpinning the research into artificial spin ice is the possibility with modern nanofabrication methods to tailor the geometry at will. This means that any arbitrary design can be realized, including more exotic arrays, not only based on extensions of the square and kagome geometries[7,14–16] but also going beyond periodic systems to artificial quasicrystals[17,18]. In addition to systems with separated elongated nanomagnets, there are also noteworthy systems with circular[19–23] or square magnets[24], or connected systems in which domain walls can travel through the network. The first efforts to create and characterize 3D artificial spin systems are also underway.

Emergent magnetic monopoles

The interest in magnetic monopoles in condensed-matter systems was sparked in 2008 when the idea of mapping the dipoles in a spin-ice system onto magnetic-charge dumbbells was proposed[25]. Soon after this work,

Key points

- Artificial spin ices are metamaterials made up of coupled nanomagnets arranged on different lattices that exhibit a number of interesting phenomena, such as emergent magnetic monopoles, collective dynamics and phase transitions.
- The motion of emergent magnetic monopoles in an artificial spin system can be controlled with external stimuli such as magnetic and electric fields, strain, temperature gradients and electric currents, which is of potential interest for future devices.
- The ability to create thermally active artificial spin ices with fluctuating moments at room temperature makes it possible to explore the rich phase diagrams with phases that are determined by the geometry, temperature and disorder.
- Signatures of the magnetic configurations are given by the specific spin-wave resonances in artificial spin ice, which offer a platform for programmable spin-wave devices, in particular magnonic crystals.
- The established artificial spin ices consist of elongated nanoscale magnets arranged on the square and the kagome lattices, but these have now diversified. New geometries include different lattices, not only periodic but also aperiodic, different magnet shapes and anisotropies, and 3D structures.
- Future work involves developments in fabrication and characterization methods, the study of artificial spin systems with new geometries and combinations of materials, and the development of applications including computation, data storage, encryption and reconfigurable microwave circuits.

the signatures of such monopoles were experimentally observed in the rare-earth titanate pyrochlores[26–28] and the term ‘magnetricity’ was coined[29], indicating that it might be possible to harness such magnetic charges in devices, just as electrons and spins are exploited in electronic and spintronic devices.

It was a natural step to consider how emergent magnetic monopoles would behave in the corresponding artificial spin ices (BOX 1). The first discussion of the theory of emergent magnetic monopoles in artificial spin ices appeared in 2009 in the context of artificial square ice[30]. However, the first experimental observations of monopole–antimonopole pairs were made in artificial kagome spin ice[5,6]. In an applied magnetic field, the monopole–antimonopole pairs were created and separated, leaving behind them 1D strings of reversed magnets, referred to as Dirac strings[6], although branching in the strings can occur[5]. The detailed behaviour during magnetization reversal is sensitive to the local energetics, which, in turn, is highly dependent on the specific geometry of the artificial spin ice[31]. Indeed, one of the appeals of artificial spin ice is that behaviour can be tuned through the lattice type and parameter, as well as the nanomagnet shape and size. Furthermore, viewing artificial spin ice in terms of emergent magnetic monopoles also provides a means to visualize the inherent physical phenomena. For example, in an artificial square ice, tracking the behaviour of magnetic monopoles during magnetization reversal gives insights into memory effects at the microscopic level[32].

The first direct observation of the spontaneous creation and separation of magnetic-charge defects, or type III vertices (see vertex types in FIG. 1c), in a thermally active artificial square ice was reported in 2013 (REF.[33]). Such defects in artificial square ice, sometimes referred to as monopoles, not only have a Coulombic interaction (like their rare-earth titanate counterparts) but also a confining potential, or string tension, binding the charge-defect pairs[30]. When an applied magnetic field is used to set the initial state of the sample with all vertices in a type II configuration, the defects separate, leaving behind strings of reversed magnets against the type II vertex background (FIG. 2a–d). However, the interaction energy between the magnetic charges, attractive or repulsive, is dependent on the background configuration[33]. In thermally active artificial kagome spin ice, the emergent magnetic monopoles remain confined, because increasing the length of the strings beyond one spin flip is energetically unfavourable[34]. Therefore, important in defining the behaviour of the monopoles are the background magnetic configuration and the artificial spin-ice geometry, both of which can be modified by introducing defects into the lattice[35–37]. In artificial square ice, the string tension can be exploited to obtain spontaneous magnetic currents, by stretching the bound monopoles apart with an applied magnetic field and removing the field in order to release them, and it was suggested that this effect may be of interest for energy storage[38]. Moreover, in the thermally active square ice, as the charge defect (type III vertex) pairs are created and spread through the system, the strings eventually coalesce to form domains of ground-state vertices separated by type II boundaries (FIG. 2e,f). Magnetic-charge defects in the form of type III vertices can propagate along the domain boundaries, providing a mechanism for the boundaries to decrease in length until they disappear[33,39] (FIG. 2g,h).

In artificial square ices, the string tension between two monopoles depends on the distance between them, but an important question is whether it is possible to create an artificial square ice in which the string tension is absent. Doing so would give truly unconfined emergent magnetic monopoles that interact via a Coulomb interaction only. The key issue is that the four nanomagnets at a vertex do not have equivalent interactions because the intermagnet separations are not all the same. One possible solution is to raise one sublattice with respect to the other[40–44], so that, at a critical vertical separation, the so-called Coulomb phase is achieved[45], with emergent magnetic monopoles freely moving in a highly degenerate, divergence-free background. Another possibility to reduce the string tension is to use two different sublattice parameters[46,47]. In lattices of connected nanomagnets, modifying the shape and size of the vertex, for example, by fabricating a hole at the vertex centre or making the nanomagnets narrower, can also make the type I and type II vertices similar in energy[48,49]. Helpful information on both the monopole motion[50] and the magnetic-charge correlation length associated with monopoles[51] can be obtained from measurements of the magnetic susceptibility. Another valuable signature of monopole defects and their Dirac strings appears in the frequency spectrum[52–54], which provides a macroscopic means of identifying monopole defects because the spectral amplitude is proportional to the number of defects. This makes artificial spin ices interesting candidates for reconfigurable or reprogrammable magnonic crystals.

Going to geometries beyond the square and kagome spin ices, a very different behaviour of the effective magnetic charges is expected[14]. For example, in the shakti

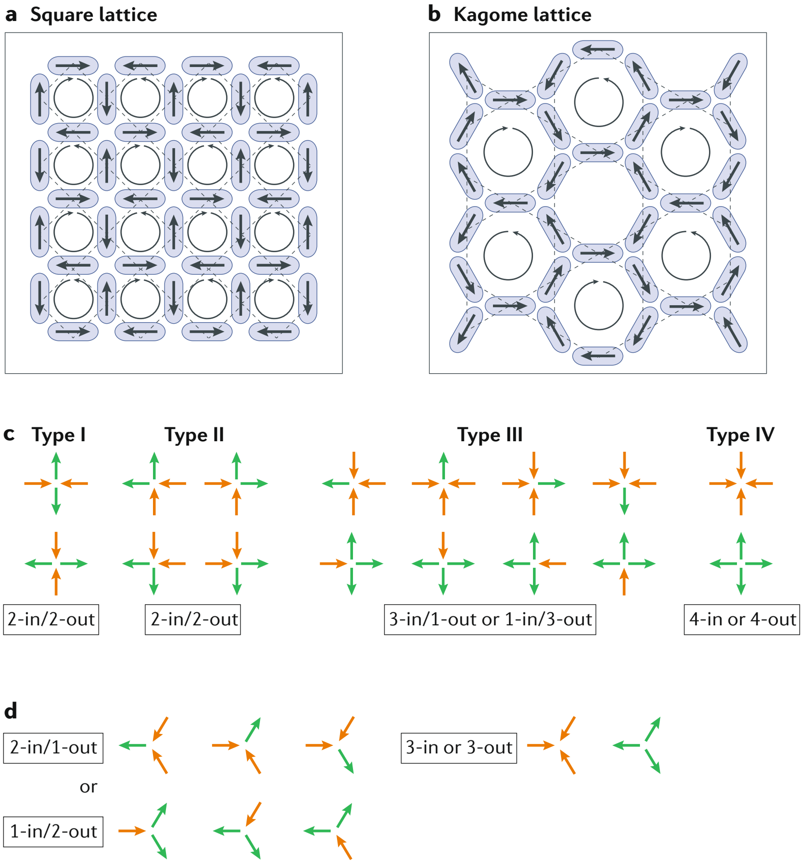


Fig. 1 | **Square and kagome artificial spin ice. a,b** | Elongated, single-domain nanomagnets are arranged on the square lattice (panel **a**) and on the kagome lattice, resulting in a honeycomb structure (panel **b**). The lattices are indicated with dashed lines as well as the ground-state configurations. The loops indicate rings of nanomagnets for which all moments are head-to-tail, resulting in flux closure. **c,d** | The possible magnetic moment configurations at the vertices for square (panel **c**) and kagome (panel **d**) artificial spin ice, with the energy of the configurations increasing from left to right.

lattice[7], it is possible to see signatures of charge screening involving magnetic-monopole polarons, analogous to polarons associated with electrons in a crystal[55]. These magnetic charges emerge at the four-magnet vertices in the shakti lattice, which are surrounded by three-magnet vertices that, because of their topology, always have magnetic charge. By means of careful design, it is also possible to control the emergent magnetic monopoles, which is important for device applications. For example, charge defects can be constrained to move in particular directions using certain geometries[56] or a temperature gradient[57], and they can also be controlled by locally inducing strain with an electric field[58].

This section would not be complete without mentioning that magnetic domain walls within the nanowires constituting connected artificial spin-ice systems can also be considered as magnetic-charge carriers[59]. These walls are chiral objects in which the magnetization within the wall rotates either clockwise or anticlockwise, independently of whether the lattice itself has chiral symmetry. Interestingly, when the walls approach a Y-shaped vertex in an artificial kagome spin ice on application of a magnetic field, this domain-wall chirality can determine whether they will move into the left or the right branch[60,61] (FIG. 2i,j). Taking advantage of this asymmetry, a domain wall passing through tailored Y-junctions, which are the basic units of connected artificial kagome spin ice, can be used to perform logic operations[62,63]. However, care is needed because the walls may transform into domain walls of opposite chirality for particular magnetic fields and geometries[64,65], and the propagation behaviour of the domain walls depends on the magnitude and orientation of the applied magnetic fields[66–68]. To circumvent these effects, the geometry of the Y-junctions can be modified so that the symmetry of the branches is reduced, giving a deterministic domain wall path that is independent of the domain-wall chirality[69]. It is also possible to nucleate domain walls in particular positions in connected artificial spin ice using a magnetic tip such as that found in a magnetic force microscope (MFM), thereby creating specific magnetic configurations at will[70]. Finally, just as the chirality of the domain walls provides additional degrees of freedom in connected artificial spin systems, the edge bending of the magnetization at the vertices in disconnected lattices also gives an additional chiral degree of freedom to the vertex monopoles[71] (FIG. 2k).

Thermodynamics and kinetics

Thermally active artificial spin systems. A fundamental question since the beginning of research into artificial spin ice has been to understand the collective behaviour of the nanomagnets, in particular, how they access thermodynamically favourable magnetic phases. The first experimental observations of a magnetic phase transition in an artificial spin system were on the square geometry, for which magnetization measurements indicated a decrease in macrospin order as the temperature was raised above a critical temperature that was below the intrinsic ordering temperature of the nanomagnet material[10]. Nevertheless, for other geometries, it has been challenging to fabricate arrays of nanomagnets that remain thermally active down to the temperatures at which their magnetic moments are predicted to thermodynamically order. In this section, we give an overview of earlier studies using thermal annealing and magnetic-field protocols, before reviewing the advances that finally led to the burgeoning work on truly thermally active artificial spin ices.

The earliest examples of artificial spin ice were athermal at room temperature, at which the experiments were conducted. Here, athermal means that the macrospins are frozen at the timescale of the measurement and, for these athermal systems, applied alternating magnetic fields were used to probe the effective thermodynamics of the system, for which an effective temperature[72,73] and effective entropy[74] were defined. True thermal ordering in these artificial spin systems was achieved by annealing the samples above the Curie temperature, T_C (REFS [75–77]). Indeed, such thermal protocols proved to be more effective for lowering the residual entropy of the system than the magnetic-field protocols used previously for athermal systems[1,78,79]. In addition to the thermal annealing of artificial spin ice followed by cooling to observe frozen-in states, it is possible to create systems that are thermally active close to room temperature.

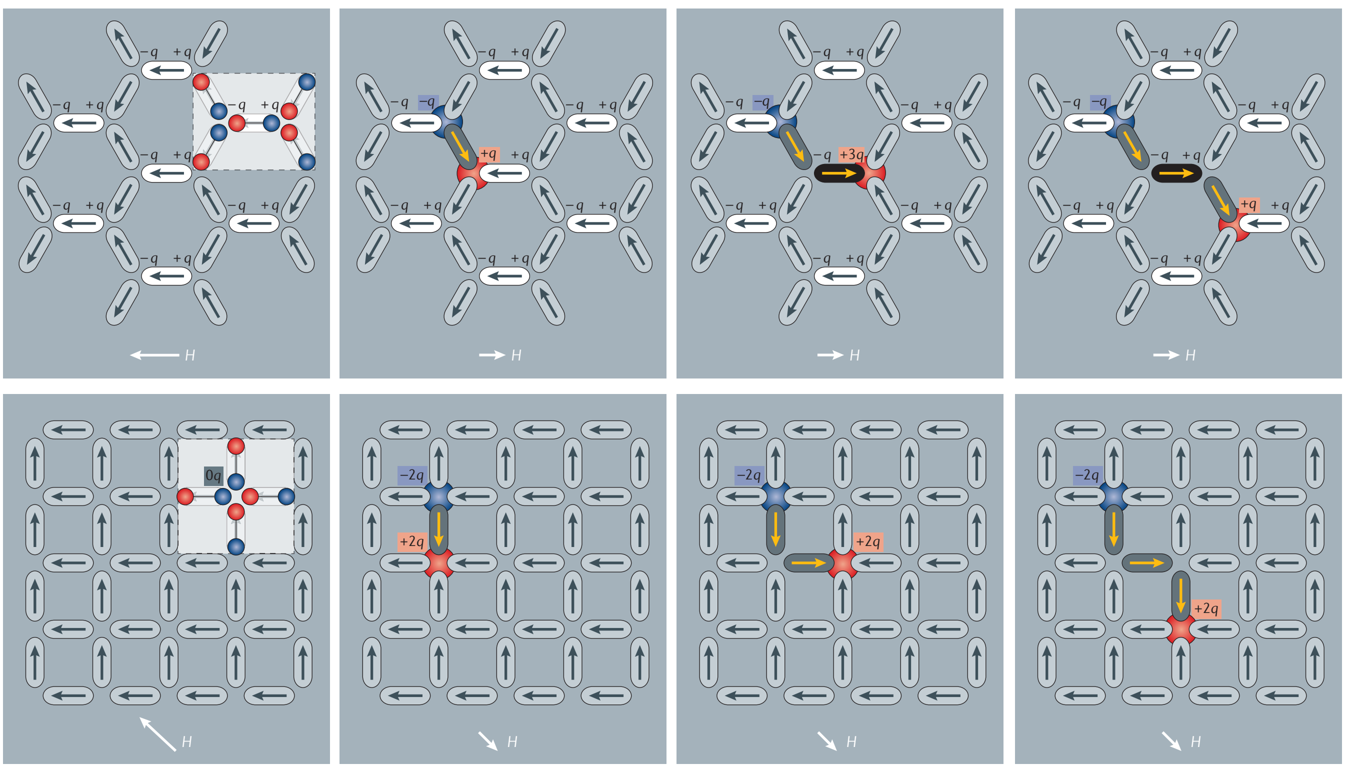
Box 1 | Emergent magnetic monopoles in artificial spin ice

We start our discussion of emergent magnetic monopoles in a system consisting of discrete mesoscopic spins in the context of artificial kagome spin ice[6,195] (see the figure, top panels). For simplicity, a reference state with all moments pointing towards the left is prepared by applying a magnetic field to the left (see the figure, top-left panel). Following the theoretical description of emergent magnetic monopoles in bulk spin crystals[25,196], each dipole moment is mapped to a charge dumbbell (see the figure, inset in top-left panel) with two opposite magnetic charges of magnitude $q = m/a$ located at its ends at two neighbouring vertices separated by a. On doing so, the net charge at the vertices alternates between $Q_0 = +q$ and $Q_0 = -q$, where Q_0 is the initial charge at a vertex, thereby forming a charge background that is non-zero at the vertices. A string of magnets can then be reversed by applying a magnetic field in the opposite direction. Whereas the vertex charge within the string is the same as the charge background, at one end of the string, its value Q exceeds the background vertex charge by $\Delta Q = Q - Q_0 = +2q$ (a monopole) and, at the other end, the background charge is reduced by $\Delta Q = Q - Q_0 = -2q$ (an antimonopole).

Thus, emergent magnetic monopoles are defined as vertices with $\Delta Q = \pm 2q$, as long as their population is dilute. However, this definition breaks down as magnetization reversal proceeds. An alternative scheme is to consider a coarse-grained charge density resulting from convolution of the vertex charge with a Gaussian of width larger than a. The regions of non-zero, coarse-grained charge density match very well with the position of the $\Delta Q = \pm 2q$ charges, and can be used to identify the emergent magnetic monopoles at any monopole concentration[6,34].

It is important to note that, during the magnet-by-magnet string expansion, the vertex charge at the ends of the string changes and the emergent magnetic monopoles are not necessarily associated with ice-rule-breaking vertices. This is an important point; the vertex charge itself is not conserved but the ΔQ charges are topologically protected quasiparticles, which means that they can only be annihilated in pairs.

Similarly, identifying vertices with $\Delta Q = \pm 2q$ makes it possible to locate monopoles in the artificial square ice (see the figure, lower panels). This situation is simpler because the magnetic-charge background is locally zero ($Q = 0$), so that the charge difference at a vertex is always equal to the vertex charge ($\Delta Q = Q$). Therefore, for artificial square ice, not only the emergent magnetic monopoles, but also their respective vertex charges, are topologically protected.



An individual nanomagnet is considered to be thermally active if the time taken for the measurement (t_m) is longer than the time required for the moment of a nanomagnet to switch (t_s), which is described by an Arrhenius law often referred to as the Néel–Brown law:

$$\frac{1}{t_s} = \frac{1}{t_f} e^{-KV/(k_B T)} \tag{1}$$

where $1/t_f$ is the attempt frequency, V is the magnetic volume of the nanomagnets, k_B is the Boltzmann constant and K is the anisotropy constant that is related to the shape of the nanomagnets and is given by $K = \frac{1}{2}\mu_0 M_S^2 \Delta N$ for ellipsoids[80], where μ_0 is the magnetic permeability of free space, M_S is the saturation magnetization and ΔN is the difference between the demagnetizing factors along the in-plane short and long axes of the nanomagnets[81]. The blocking temperature (T_b) is defined as the lowest temperature at which the nanomagnets appear to be thermally active, or superparamagnetic (or perhaps more correctly 'paramagnetic superspins'), at the timescale of the measurement (that is, $t_s < t_m$). Below T_b, the average time taken for moments to switch is much longer than the measurement time, and the nanomagnets appear to be frozen. For the nanomagnets in artificial spin ice, the switching can occur through a non-uniform process[82], leading to a much lower T_b than is expected for magnets displaying single-domain coherent rotation. Equation 1 still applies, but the energy barrier is modified to replace the true volume of the nanomagnet with a smaller effective activation volume. Interestingly, the moments of an artificial spin ice comprising weakly coupled nanomagnets freeze in a similar manner to that of a glass upon approaching T_b from above; at T_b, the system crosses over from the thermally equilibrated state to the frozen state following a Vogel–Fulcher–Tammann law[82,83].

To observe thermally active behaviour in artificial spin ice, the blocking temperature T_b must be below the temperature of the measurements, which, for practical reasons, are ideally performed close to room temperature. Equation 1 provides some insight into how to lower the blocking temperature T_b by modifying the individual nanomagnets, despite the fact that it does not fully reflect their switching mechanism. In particular, the volume of a nanomagnet can be decreased by reducing its thickness and/or the lateral dimensions[33,84,85], and the shape anisotropy of a magnet can be reduced by lowering its aspect ratio (nanomagnet length relative to width) or magnetization. Typically, the nanomagnets of artificial spin ice are fabricated from Permalloy, a soft magnetic alloy that has a composition close to 80% Ni/20% Fe. Therefore, to reduce the magnetization of the Permalloy, the Ni content can be increased, thus lowering the T_C of the material[75]. Another possibility to reduce the magnetization is to alloy magnetic elements with non-magnetic elements, such as Pd (REFS[77,86]), or to use a ferrimagnetic material instead of a ferromagnetic material[87].

Beyond observing thermally active behaviour, it is also desirable to observe magnetic ordering in artificial spin ices. For this, the nanomagnets need to be thermally active below the temperature of the phase transition of interest. However, all of the routes to lower T_b that involve decreasing the nanomagnet moment, $m = M_S V$, also simultaneously lower the phase-transition temperature, because the dipolar coupling between the nanomagnets, which is proportional to m^2, is reduced. To counteract this effect on the transition temperature, and ensure it is above T_b, the lattice parameter can be minimized[76,85]. This decreases the nanomagnet separation and increases the coupling between the nanomagnets, which is also proportional to d^{-3}, where d is the nanomagnet centre-to-centre distance. Therefore, great care in the design of artificial spin systems is required to ensure that the system reaches equilibrium. Also important is the choice of thermal protocol. A useful discussion on determining whether a system is out-of-equilibrium can be found in a recent colloquium[88].

Phases and phase transitions. Progress in fabrication techniques, which has allowed the manufacture of artificial spin systems with spontaneously fluctuating moments, has fuelled the research into the collective behaviour of thermally active artificial spin systems. We review the latest findings regarding the prototypical square and kagome lattices, before touching on the more exotic and still rather unexplored lattices.

The predicted ground state in artificial square ice[42] was experimentally verified in 2011 in samples that ordered during the early stages of deposition of the Permalloy film from which they were made[89]. The ordering was later reproduced using thermal annealing protocols[75–77] and observed in thermally active systems[33]. This ordering results from the fact that, although all ice-rule-obeying vertex configurations have a low energy, the non-equal interactions between the nanomagnets at the vertices lift the degeneracy and favour type I vertices[1]. Therefore, artificial square ice does not have an extensive degeneracy, and it is of interest to clarify the nature of the phase transition to the low-energy state[90,91]. Indeed, in recent X-ray scattering experiments, first measurements of critical behaviour across the phase transition have been obtained, providing evidence of a continuous transition belonging to the two-dimensional Ising universality class[12]. In addition, it may be possible to alter the nature of the phase transition by modifying the energies of the vertex configurations, for example by using an exchange-bias field[92].

One of the several ways proposed to equalize the intermagnet interactions in the square ice geometry is to lift the nanomagnets on one of the two sublattices out of the plane[40–42] (FIG. 3a), an approach which has been experimentally realized[43,44]. Whereas the structure factor deduced from the MFM images of conventional square ice displays the expected peaks associated with

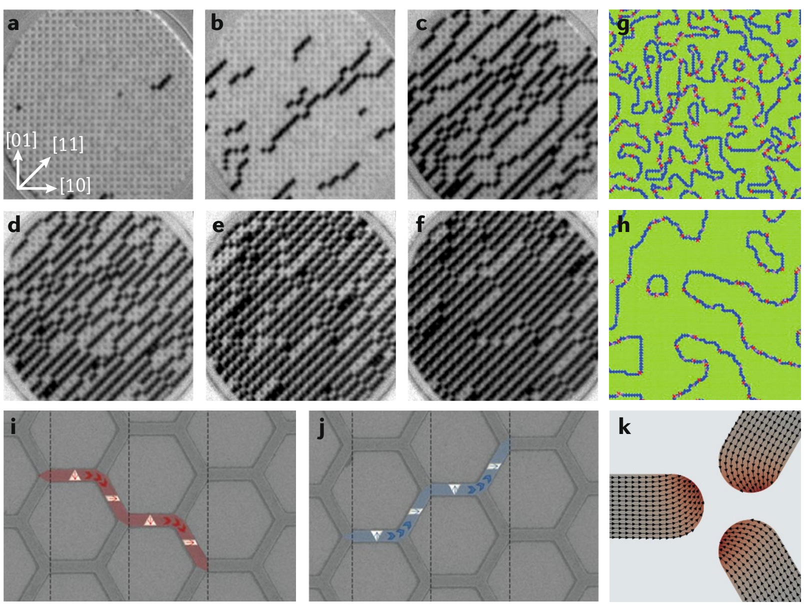


Fig. 2 | **Propagation and chirality of monopole-like magnetic charges in artificial square and kagome ice. a–f** | Observation of thermal relaxation in artificial square ice. Initially, an array of type II vertices is created after application of a magnetic field along the negative [11] direction, ensuring that all moments point towards one direction. Magnetic-charge defect pairs are created and separate, leaving behind strings of reversed magnets, which eventually coalesce to form ground-state domains separated by domain boundaries. The field of view is 20 µm. **g,h** | The domain boundaries themselves contain magnetic-charge defects, whose motion results in a decrease in the length of the domain walls that eventually annihilate. **i,j** | Domain walls within the magnetic elements of a connected artificial kagome spin ice have a chirality; a down or up orientation of the magnetization within the wall influences its path through the nanowire network. **k** | Monopoles with an additional degree of freedom, arising from the chiral nature of edge bending at a vertex. The distance from the tip of the nanomagnets to the centre of the vertex is 32 nm. Panels **a–f** are reprinted with permission from REF.[33], APS. Panels **g** and **h** are from REF.[39], © Deutsche Physikalische Gesellschaft. Reproduced by permission of IOP Publishing. CC BY-NC-SA. Panels **i** and **j** are reproduced from REF.[60], CC-BY-3.0 https://creativecommons.org/licenses/by/3.0/. Panel **k** is from REF.[71], © Deutsche Physikalische Gesellschaft. Reproduced by permission of IOP Publishing. CC BY-NC-SA.

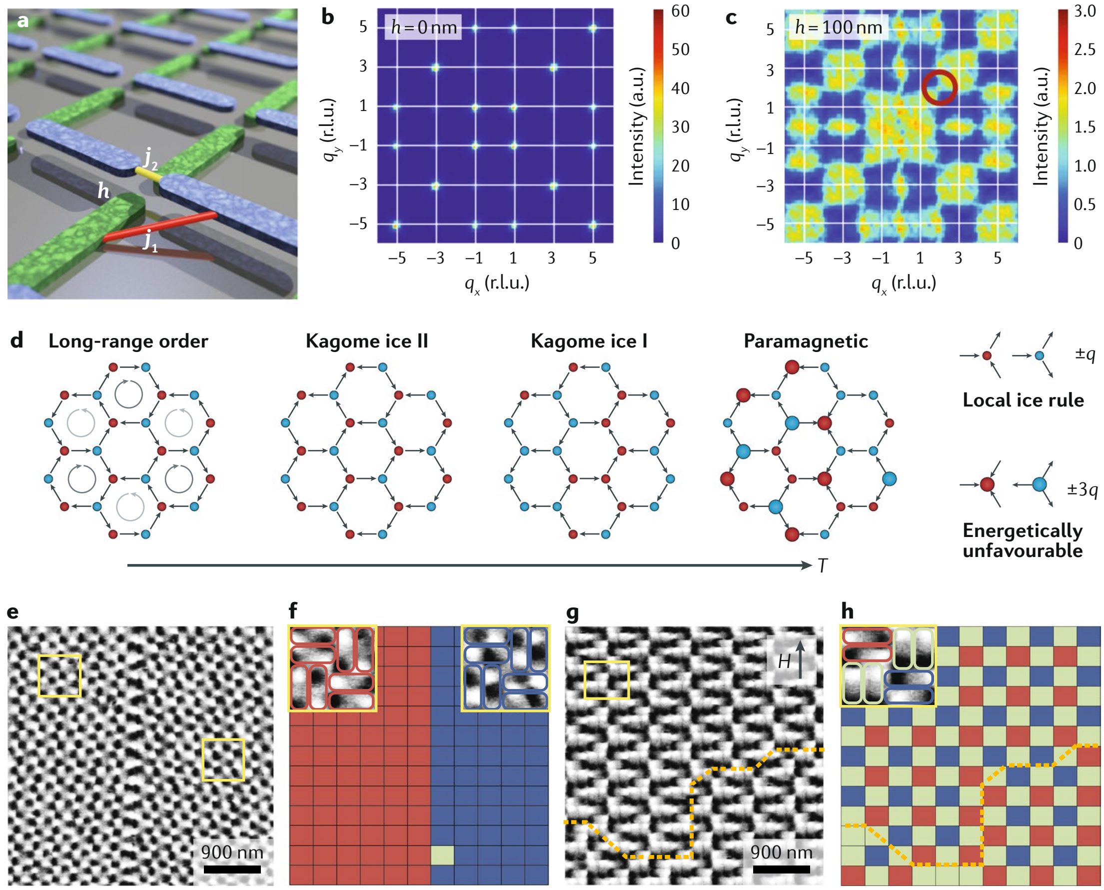


Fig. 3 | **Magnetic order and phase transitions. a** | Schematic of an artificial square ice in which one sublattice is offset out of the plane by a height h to modify the j_1 interaction strength between nearest-neighbour nanomagnets relative to the j_2 interaction between next-nearest neighbours. **b**,**c** | Magnetic-structure factors deduced from magnetic force microscopy (MFM) images of the artificial square ice schematized in panel **a**, for $h = 0$ nm (panel **b**) and $h = 100$ nm (panel **c**). The colour scale refers to the intensity at a given location (q_x, q_y) in reciprocal space. The red circle in panel **c** encloses a pinch point. **d** | Schematics of the predicted magnetic phases as a function of temperature (T) in artificial kagome spin ice[42]. Magnetic charges of opposite sign are indicated in red and blue at the vertices, with small dots denoting $\pm q$ and large dots denoting $\pm 3q$ magnetic charges. **e**–**h** | Ordered Potts states in a 'quadrupolar' ice with plaquettes of pairs of nanomagnets. MFM image (panel **e**) and corresponding bitmap (panel **f**) of a zero-field-annealed system favouring a doubly degenerate ferro-quadrupolar phase because of strong intraplaquette dipolar interactions. MFM image (panel **g**) and corresponding bitmap (panel **h**) of a lattice annealed in a magnetic field, which favours an antiferro-quadrupolar phase, because of strong interplaquette dipolar interactions. The boundary between the two antiferro-quadrupolar domains in panels **g** and **h** is indicated by the yellow dashed line. The yellow square frames each contain four plaquettes of moment pairs; the colours in the bitmaps represent the different Potts states for the moment pairs, where red and blue denote the ferro-quadrupolar states with opposite chirality (+1 and −1) and green denotes the antiferro-quadrupolar state. r.l.u., reciprocal lattice units. Panels **a**–**c** are reprinted from REF.[43], Springer Nature Limited. Panels **e**–**h** are adapted from REF.[108], Springer Nature Limited.

ordering (FIG. 3b), 'pinch points' appear in the structure factor for the fully degenerate square ice (FIG. 3c). These pinch points are a signature of a so-called Coulomb phase[45], which obeys Gauss' law and supports magnetic-monopole excitations that emerge in highly frustrated systems. The same effect has been demonstrated in simulations of a connected 2D square ice by tuning the size of the cross-shaped vertices[48].

It is predicted that the artificial kagome spin ice, despite the extensively degenerate configurations arising from the geometrical frustration, has a rich phase diagram (FIG. 3d), including a long-range-ordered phase, because of the dipolar interactions between the nanomagnets[42,93]. On cooling, there is first a crossover from a disordered paramagnetic phase to another paramagnetic phase, referred to as kagome ice I, in which the ice rule is obeyed with magnetic charges of $+q$ or $-q$ at each vertex. The cooperative paramagnetic behaviour of the kagome ice I phase means that it can be seen in the framework of spin-liquid physics[88]. A spin liquid is a magnetic state that is disordered but correlated, and the interactions within the assembly of classical Ising variables lead to non-zero pairwise spin correlations, which decay to zero at large distances.

Upon lowering the temperature further, the system passes through two phase transitions; first, an Ising transition to the kagome ice II phase with ordered magnetic charges, but still with highly fluctuating moments, and then a transition to a long-range-ordered phase with both charge and spin order. This last transition is a second-order three-state Potts transition, although simulations give a Kosterlitz–Thouless transition if charge-order-breaking fluctuations are not present[93,94]. Large system sizes are needed to observe the three-state Potts universality because these charge fluctuations are rare. Intriguingly, the emergence of the charge-ordered phase is not easily explained, as no charge degree of freedom is encoded in the underlying dipolar spin Hamiltonian[88,95]. However, the emergence of the charge order and the rest of the phase diagram can be understood in terms of fragmentation of the spins into two parts; a part with maximum divergence in the magnetic field, which gives a static signature of ordered magnetic charges, and a divergence-free part, reflecting a disordered magnetic phase with fluctuating moments characteristic of a Coulomb phase[95–97]. The kagome ice II phase can thus be regarded as a spin liquid superimposed on a magnetic-charge crystal.

The difficulty in obtaining ordering in thermally active artificial kagome spin-ice structures with sizes beyond four hexagonal rings has been demonstrated[84,98]. In extended kagome lattices, an indication of two phase transitions has been observed with low-energy muon spin spectroscopy[11], and the first signatures of the relevant phases have been observed with neutron scattering and resistivity measurements performed on connected artificial kagome spin ice[99–102]. However, only small areas of the charge-ordered phase have been directly observed with imaging techniques following annealing[76,77,103], and the long-range ordering of the magnetic charges (kagome ice II phase) or the spins (the ground state) have not yet been unambiguously observed. The difficulty in achieving ordering in the artificial kagome spin ice results from the high frustration and extensive degeneracy of the system. Achieving the ground state in such a degenerate energy landscape simply takes a very long time and may also be hindered by kinetic restrictions. For example, during ordering of the magnetic charges into the kagome ice II phase, domains with an even number of vertices are favoured. This observation indicates that two vertices with opposite charge prefer to order simultaneously so that charge neutrality is maintained, thus creating restrictions on how the domains grow[87]. Another challenge associated with achieving the ground state through thermal relaxation in highly frustrated systems is that such systems are susceptible to residual magnetic fields, which can dramatically change the energy landscape[104].

There are many more geometries beyond the kagome and square geometries. For more exotic lattices, it was found that the low-energy states adopted by an extended system could not be predicted by only considering the relative energies of the building blocks consisting of a few nanomagnets[18,105]. Going beyond Ising-like systems, dipolar XY systems with arrays of coupled discs exhibit complex ordering phenomena[19,20,22], and the phase diagrams of such XY systems can be modified by structural disorder to give new phases that favour local flux closure[20]. Both the lattice and the moment degrees of freedom define the ordering behaviour. For example, the kagome Ising system with out-of-plane moments[88,106,107] displays behaviour that is quite different from the kagome lattice with in-plane moments. Furthermore, quadrupolar ice exhibits competing ferromagnetic and antiferromagnetic orders[108]. Here (FIG. 3e–h), pairs of elongated nanomagnets, which can be considered to be quadrupoles, are arranged in a similar manner to the individual nanomagnets in the so-called chiral ice. The quadrupolar ice provides a realization of the Potts model, which can also be achieved by exploiting magnetocrystalline anisotropy in epitaxial thin films[24], with competing interactions in the system resulting in a complex phase diagram that is dependent on the magnetic field and temperature.

Fast dynamics

Resonances and spin-wave excitations. Because the nanomagnets in artificial spin ices are typically manufactured from soft magnetic metals such as Permalloy, the magnetic resonances and spin-wave excitations typically investigated occur in the 1–100-GHz range. When the artificial spin ices are made up of disconnected nanomagnets, the observable spin-wave excitations comprise confined modes excited within each individual nanomagnet and can be thought of as modes of precession that can be strongly affected by the nanomagnet shape, applied magnetic-field direction and stray field from neighbouring nanomagnets[109]. For example, excitation spectra obtained from an artificial square-ice array of nanomagnets can be directly mapped onto the spectra expected from a single, isolated rectangular nanomagnet that supports standing and localized modes, and the behaviour of the frequencies of these modes in an applied magnetic field are well understood[110] (FIG. 4a). Because of this one-to-one mapping, probing spin-wave resonances for a static magnetic field applied at different angles gives valuable information that can be readily interpreted in terms of the nanomagnet properties[54], allowing the identification of resonances from sets of nanomagnets with particular orientations, for example.

Collective excitations of the nanomagnets in artificial spin ice have been predicted and studied for several geometries. In particular, the relationship between the lattice geometry, the magnetic configuration present and the resonant modes that occur within the nanomagnets has been demonstrated with experimental measurements on artificial square ice, which revealed multiple modes in the spectra in addition to the main resonance[111]. Micromagnetic simulations[112] and scaling arguments[51] show that the modes can be understood in terms of the number and location of emergent magnetic monopoles. In particular, the emergent magnetic monopoles created during magnetization reversal in an applied magnetic field (FIG. 4b) can be detected using ferromagnetic resonance (FMR) that, with its sensitivity to local magnetic fields, provides a way to distinguish contributions from different vertex types in the spectra[113,114]. The frequency spectrum for an artificial square ice in the low-field regime (FIG. 4c) reflects the generation of pairs of oppositely charged type III vertices separated by a string of magnets with reversed moments[52] (FIG. 4b). From these spectra, it is possible to identify features related to the string length and the number of defect pairs. The amplitudes of the spectra, for example, are

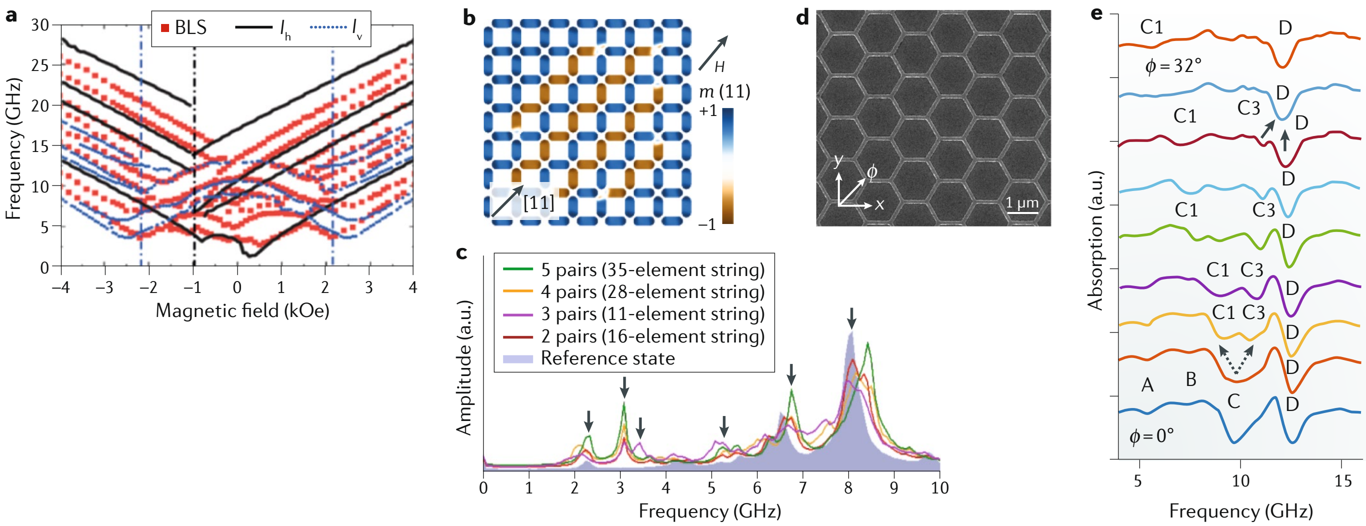


Fig. 4 | **Fast dynamics in square and kagome artificial spin ices. a** | Frequencies as a function of magnetic field measured with Brillouin light scattering (BLS), with simulation results for isolated nanomagnets that are horizontal (I_h) and vertical (I_v) with respect to the field. The vertical lines indicate the switching fields of the nanomagnets. **b** | Artificial square ice with 112 nanomagnets, used for simulating spin-wave spectra for an applied magnetic field in the (11) direction. The lattice contains four monopole–antimonopole pairs connected by Dirac strings. **c** | Spin-wave spectra calculated for strings of reversed macrospins in the lattice shown in panel **b** for increasing string length and number of monopole–antimonopole pairs compared to the reference state. The arrows indicate the main, distinct signatures of the topological defects. The resonance peaks shift to higher frequencies when the strings become longer. **d** | Scanning electron micrograph of a connected artificial kagome spin ice. **e** | Resonance response of the connected artificial kagome spin ice for different directions φ of the applied magnetic field ($H = 1$ kOe). At $\varphi = 0°$, four modes are observed (A, B, C and D). C splits into two modes (C1 and C3) when φ increases, and C3 merges with D at $\varphi \approx 30°$. The three modes C1, C3 and D originate from nanomagnets with different orientations. Panels **a** and **b** are adapted from REF.[110], Li, Y. et al. Thickness dependence of spin wave excitations in an artificial square spin ice-like geometry. *J. Appl. Phys.* **121**, 103903, https://doi.org/10.1063/1.4978315 (2017), with the permission of AIP Publishing. Panels **c** and **d** are adapted with permission from REF.[52], APS. Panels **e**–**g** are adapted with permission from REF.[54], APS.

related to the length of strings, whereas the frequencies are related to the location of the defects relative to the nanomagnet-array boundaries. The magnetic configurations therefore have telltale signatures associated with the distribution of local magnetic fields acting on the associated nanomagnets[52].

Signatures in the resonant spectra can also arise from the non-uniformity of magnetization within the individual nanomagnets. In particular, sufficiently large nanomagnets minimize their stray-field energies by canting spins near the nanomagnet ends[115]. This canting modifies the energy of the resonances because it enables additional configurations at the vertices. For example, the degeneracy of the ground state in an artificial square ice (FIG. 1a) can be lifted, resulting in two peaks where otherwise there might be only one[115].

Investigations of spin-wave resonances in frustrated, artificial spin systems are not restricted to Ising-like systems. For example, coupled magnetic vortices arranged in a kagome geometry[116] display collective oscillation modes of the interacting vortex cores that are sensitive to an applied magnetic field. This system is particularly interesting because frustration leads to multiple vortex configurations, reflected by the resonant frequencies.

Magnonic crystals. In addition to using spin waves as a means to study defect formation in artificial spin systems, artificial spin-ice geometries may offer a platform for programmable spin-wave devices[3]. In this regard, detection of spin waves in artificial spin ice with spin-torque spectroscopy may be especially useful[117]. A connected geometry able to transmit spin-wave information through the lattice is particularly suited to the control of the manifold of spin-wave states[118]. This concept is widely used as the basis for control of optical excitations in photonic crystals designed to diffract optical waves at certain wavelengths by means of patterned structures. The analogous, patterned materials for spin waves are referred to as magnonic crystals and metamaterials, and there is a large body of work on the topic of magnonics and magnonic devices, for which composite and 3D materials are possible future directions[13].

The concept of controlling spin-wave band gaps and properties by adjusting interactions between the magnetic components is being explored[119–121]. In artificial spin ice, this control can be achieved with end canting[115] and by choosing interfaces designed to induce chiral-symmetry-breaking Dzyaloshinskii–Moriya interactions. In addition, a periodic, artificial spin-ice geometry can be used to create a spin-wave band structure in which the frequencies near the zone centre can be tuned by setting the magnetic configuration[122]. The introduction of Dzyaloshinskii–Moriya chiral symmetry breaking in artificial spin ice is also of interest for magnonic devices, giving nonreciprocity as well as topologically protected spin waves in the long-wavelength limit[123].

Difficulties with using nanometre-scale magnetic structures for magnonic applications can arise simply because the material volumes are small. The small volumes can result in weak coupling between nanomagnets and signal-to-noise issues. Therefore, the material volume can be maximized by creating an array of holes in a magnetic thin film[124], a system often referred to as an antidot array[125]. The vacancies can be left empty or filled with another material[126]. Depending on their shape, the vacancies create specific patterns in the local magnetic fields generated by the surrounding material and, thus, define the magnetic environment that controls spin-wave propagation. For example, an array of elliptical holes arranged in an artificial square-ice geometry supports a magnonic band structure as well as a type of spin-wave channelling, as characterized with Brillouin light scattering[127]. Another key aspect of such connected geometries is the possibility to modify and enhance spin waves through the junctions connecting the nanomagnets[13,128]. For example, in connected artificial kagome spin ice[54], strong signatures from monopole resonances, shifted compared with those of unconnected kagome ice, can be obtained for a two-step magnetization-reversal process driven by a magnetic field applied along different lattice directions (FIG. 4d,e).

Geometries and associated phenomena

One of the main attractions of the artificial spin-ice approach is that novel phenomena can be observed in nanomagnet arrays for which every aspect can be engineered with various nanofabrication methods. In this section, we discuss a broad range of artificial spin systems — many of which go beyond artificial spin ice — that all share the philosophy of generating new, emergent phenomena that arise from the collective behaviour of constituent building blocks. In this sense, artificial spin ices are examples of metamaterials.

A variety of geometries of artificial spin systems have been implemented to date; these geometries and their relationships are shown in BOX 2. Indeed, the only limitation on possible designs is the imagination of the designer. Even just a small change, such as the introduction of lattice defects, can make an important difference to the behaviour. Lattice defects are topological defects in the structure (FIG. 5a), and the introduction of an edge dislocation in the artificial square ice results in the creation of domain boundaries between areas of type I ground-state order that have one end pinned at the defects[37].

One of the first alternative lattices to the kagome and square geometries was the 'brickwork' lattice[129], which keeps the long axes of the nanomagnets orthogonal to each other, just as for square ice, but reduces the number of nanomagnets meeting at each vertex to three, making it topologically equivalent to the artificial kagome spin ice. However, because the lattice is not highly frustrated, its properties are more similar to the square than the kagome artificial spin ice. This result demonstrates that the details of the geometry and symmetry are important in artificial spin systems, and topology alone is not enough to classify such systems.

The four vertex types of the standard square ice have a particular hierarchy of energies (FIG. 1c). This hierarchy can be modified to increase the frustration of the macrospins on the lattice. There are several approaches that, compared to offsetting one of the sublattices out of the plane (FIG. 3a), are easier in terms of the fabrication. For example, stretching the lattice to a rectangular shape increases the degeneracy of vertex states[46,47]. In another example, introducing magnetic nanodiscs into the gaps at the vertices[21], as so-called 'slave' macrospins, equalizes the energy between type I and type II vertices. The macrospin of a disc is XY-like, taking on any in-plane direction, but as a 'slave' in the artificial square ice, its direction is dictated by its magnetostatic interactions with the four elongated nanomagnets that surround it at the vertex. At a critical disc size, there is no longer an energy gap between the type I and type II vertex configurations, and pinch points are observed in the magnetic-structure factor, indicating that the degeneracy of the ice-rule-obeying vertices has been restored. It is also possible to tune vertex energies by modifying the vertex shape and size in systems of connected elements by inserting a hole at the centre of the vertex[49] or decreasing the width of the elements[48].

Another system based on the square ice for which the vertex-configuration energies can be tuned is the dipolar trident system. In this system, each element in the square ice is reproduced to give a group of three parallel nanomagnets, which then interact among themselves[130]. These interactions modify the energies of the different vertex types, and the vertex energies can be adjusted by modifying the distances between the three elements in the group and the distances between the groups.

Chirality is the basis for several interesting effects in artificial spin systems. The trident lattice can have an inherent chirality that arises from the geometry of the trident design[130]. In addition, a dynamic chirality was observed in the pinwheel system[131,132], referred to as chiral ice[9], in which the nanomagnets in a square ice are rotated through 45° (FIG. 5b). In this case, minimization of the stray-field energy at the edges of the array results in an unexpected ratchet effect, in which thermal relaxation leads to a rotation of the magnetization of the array in a unique direction defined by the edge structure. In addition, the chiral ice prefers specific ferromagnetic ordering upon tuning the anisotropy of the system and the topology of its edges[131], and this leads to the appearance of flux-closure states similar to those found in mesoscale thin-film-patterned elements. Further geometries incorporating chiral plaquettes are the square–kite tessellation[105] and the vortex ice, which is based on the artificial kagome spin ice and results in a defined chirality of the vortex states in the hexagonal rings of nanomagnets[133]. Handedness in the magnetic order has also been observed in an artificial magneto-toroidal crystal, for which the toroidal moment can be defined with the magnetic tip of an MFM[134]. In this geometry, the nanomagnets are arranged in squares that are repeated across the array, which is equivalent to replacing each element in artificial square ice with two elements (see also REF.[124] and supplementary information in REF.[135]). Hence, the lattice geometry is non-chiral, but the lowest-energy magnetic configuration is chiral. Similarly, edge bending occurring at monopoles with magnetic charge $\pm 3q$

in the kagome lattice (FIG. 2k) adds magnetic chirality to an otherwise non-chiral lattice geometry.

It is possible to make different vertices play different roles by designing them to have different coordination numbers. An example of this class of lattices with mixed coordination numbers is the pentagonal lattice[55], because it has either four or three nanomagnets meeting at each vertex. Mixed-coordination-number lattices

Box 2 | **The family of artificial spin systems beyond artificial spin ice**

Here, we highlight some of the connections between the different lattices that have either been realized experimentally or considered theoretically. The square ice[1] and the kagome ice[42] (highlighted figure parts) are the prototypical 2D lattices of elongated nanomagnets with in-plane magnetization, and variants of these have been achieved either by adding, removing, rotating or elevating nanomagnets systematically, or by combining lattices.

Rotating the nanomagnets in the square ice by 45° leads to the chiral ice, which is also called the pinwheel lattice[9,131]. Placing two parallel nanomagnets at each site of the square lattice, instead of one, leads to the toroidal lattice[134]. Rotating the nanomagnets in the toroidal lattice by 45° leads to the quadrupolar lattice[108]. Adding yet another magnet to each site in the quadrupolar lattice leads to the trident lattice[130]. The chiral plaquettes of four nanomagnets in the chiral ice can also be combined differently to give the square–kite tessellation lattice[105]. Decimating the square lattice in various ways leads to the tetris[15], shakti[14,15] and brickwork[197] lattices. A quasi-3D structure can be obtained by offsetting one sublattice of the artificial square ice out of the plane[43]. The square ice can also be combined with another square lattice of circular nanodiscs[19,22] to give the square ice with XY modifiers[21].

Two overlapping kagome lattices for which one is shifted with respect to the other give the dice lattice[16]. Adding extra vertices between kagome rings leads to the vortex lattice[133] and stretching the kagome rings in one direction and adding nanomagnets gives the pentagonal lattice[55]. 2D lattices with circular disc magnets can be of square[19,22], triangle[22], kagome[22] and honeycomb[22] geometries.

In further 2D systems, the nanomagnets can be placed on the shuriken[198], modified shakti[137], Santa Fe[8], Ising stripe[199] and rectangular[47] lattices.

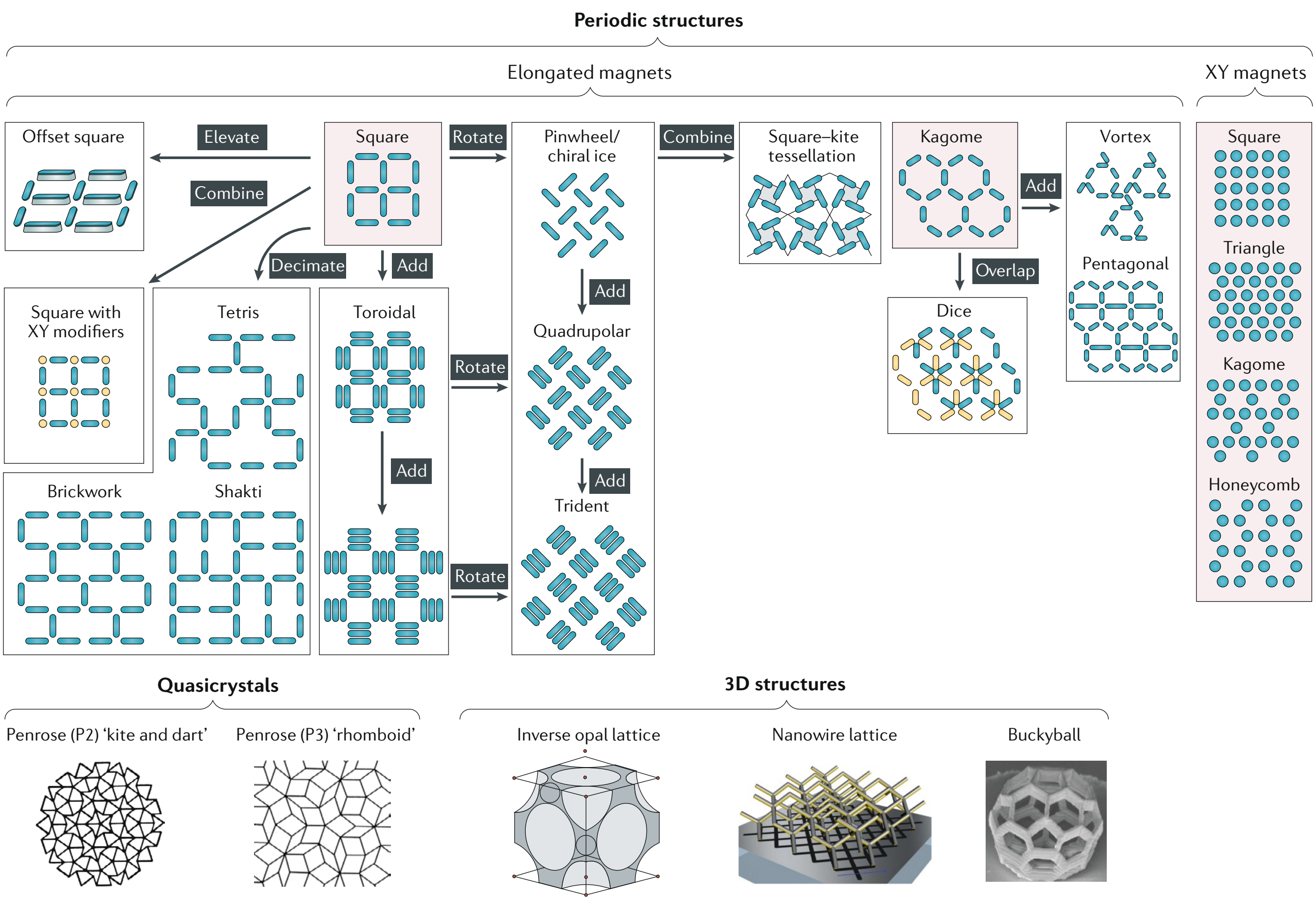
The in-plane magnetization of the nanomagnets can also be constrained by introducing magnetocrystalline anisotropy, giving a realization of the Potts model[24].

Going beyond the periodic lattices, we find the quasicrystal lattices of connected, elongated elements, with the prototypical 'kite and dart'[17] and 'rhomboid'[18,145] geometries. 3D structures include the inverse opal[153,200] and nanowire[161] lattices, and the buckyball[162].

Another class of systems consists of magnetic dots with out-of-plane magnetization, which can be fabricated with the same lattice geometries as the XY systems (highlighted figure part), although their behaviour is fundamentally different[106,201,202].

For the quasicrystal lattices, we highlight connected geometries, although quasicrystal systems made up of individual magnets have also been investigated. In addition to the connected quasicrystal lattices, examples of other lattices with connected elements exist in, for example, square[117], kagome[54] and brickwork[197] geometries. These are topologically equivalent to antidot lattices[125] that are interesting as magnonic crystals and as networks for the observation of domain-wall motion.

Penrose (P2) 'kite and dart' image is adapted with permission from REF.[17], APS. Penrose (P3) 'rhomboid' image is adapted with permission from REF.[145], APS. Inverse opal lattice image is reprinted with permission from REF.[200], Liu, Y. et al. Confined chemical fluid deposition of ferromagnetic metalattices. *Nano Lett.* **18**, 546–552 © 2018, American Chemical Society. Nanowire lattice image is reproduced from REF.[161], CC-BY-4.0 https://creativecommons.org/licenses/by/4.0/. Buckyball image is reprinted with permission from REF.[162], APS.



not only house emergent magnetic monopoles but also associated emergent magnetic polarons. Such screened magnetic charges have also been directly visualized as a transient state in the dice-lattice artificial spin ice, which has vertices with coordination of three or six[16].

Allowing mixed coordination numbers permits the construction of many more lattices, including the shakti, tetris and Santa Fe lattices[8]. The most heavily studied of these is the so-called shakti lattice, constructed by removing elements from the square-ice system in such a way that there is a mixture of coordination numbers 3 and 4 (FIG. 5c). This decimated square ice has been shown theoretically to possess a quasicritical ice phase with critical correlations similar to those found in the Coulomb phase of the pyrochlore spin ices[136]. Artificial spin ices based on the shakti lattice have been experimentally realized in both athermal[7] and thermally active[137] forms. In this class of vertex models, the frustration does not arise from the properties of an individual vertex but, rather, from the inability of neighbouring vertices to take up their lowest-energy configuration at the same time[138]. The phenomenon is known as topologically induced emergent frustration[7], and such 'unhappy vertices' thus provide topologically protected excitations above the ground state. In proper geometries, the degeneracy of the allocation of such vertices grows exponentially with the size of the system, leading to a degenerate low-energy manifold[8,139]. This vertex frustration can only be realized with artificial spin systems because they have no known analogues in nature, and remains a fruitful avenue for future work. A modified version of the shakti lattice in which some of the nanomagnets are longer than the others has been shown to order differently, because of the difference in the energy barriers to switching for the different nanomagnets[137]. Combining designs with different density and connectivity, such as in the tetris lattice[15], can also give rise to regions with different energy barriers to switching, as well as regions that display magnetic behaviour that is expected for 1D systems.

In addition to using elongated magnets for Ising-like systems, it is possible use circular nanomagnets to create so-called dipolar-coupled XY systems with stripe-ordered low-energy states[19,22]. Introducing positional disorder in such artificial spin systems modifies the phase diagram to give a microvortex phase[20]. Introducing random interactions and/or removing crystalline order can be a route to obtain spin-glass behaviour[140], which has been investigated in dipolar XY systems with positional disorder[20] and systems of randomly placed elongated magnets[141]. In addition to using shape anisotropy to constrain the magnetic moments, it is possible to use single-crystal materials to define the in-plane magnetization direction, restricting the nanomagnet moments to point along the crystallographic axes. For example, a tunable, dipolar, four-state Potts model system can be created employing single-crystal nanomagnets with cubic anisotropy, which can order ferromagnetically (FIG. 3e,f), antiferromagnetically (FIG. 3g,h) or in an ice-like manner, depending on the relative orientation between the easy axes and the bonds connecting the magnets[24].

Between crystals, with perfect discrete translation symmetry, and glasses, with total absence of any such symmetry, lie the quasicrystals. These possess order — knowledge of a small part of the structure and a short list of rules is enough to construct the structure over all of space — but lack discrete translational symmetry. Bulk magnetic quasicrystals include both rare-earth[142] and transition-metal[143] magnetic species and display spin-glass-like freezing when the spins are dilute. A 2D analogue of a quasicrystal is the Penrose tiling[144]. Indeed, Heisenberg spins on the nodes of a Penrose tiling have been studied theoretically and recreated as a system of macroscopic magnets[145], and the resulting low-energy magnetic configuration corresponds to interpenetrating, non-collinear sublattices in a higher-dimensional structure[146]. This work provided the inspiration to build a microscopic artificial analogue of a magnetic quasicrystal by placing nanomagnets with Ising-like macrospins along the edges of a Penrose tiling.

Penrose patterns come in two forms, known as kite and dart or rhomboid, with the names derived from the shapes of the tiles used to form them. Permalloy kite-and-dart lattices, in which the magnetic elements are connected to form a continuous network, have been investigated using the macroscopic probes of SQUID magnetometry and FMR, revealing well-defined switching fields and a rich mode structure[147]. Magnetic imaging of such patterns reveals spatially distinct ordered and frustrated sublattices[17] and reversal by means of 2D avalanches[148], with vortices forming metastable transient states[149]. An artificial magnetic quasicrystal built by placing discrete nanomagnets along the edges of rhomboid tiles[18] (FIG. 5d), allows the effects of coupling to be studied by varying the spacing between the nanomagnets. In contrast to the kite-and-dart quasicrystal, all the links between vertices of the rhomboid tilings are the same length, and this lattice type is less frustrated than the kite-and-dart lattice[150]. Theoretical considerations and MFM imaging show that the pattern contains a rigid skeleton of moments that spans the pattern and possesses a two-fold degenerate long-range-ordered ground state. The ground state surrounds moments that can be switched without affecting the total energy of the system, leading to an extensive degeneracy. This behaviour is reminiscent of the heterogeneous ordering in the tetris lattice[15]. This behaviour is also similar to the

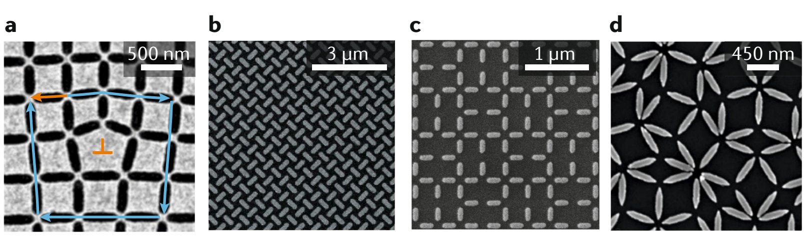


Fig. 5 | **Experimental realizations of selected artificial spin-ice geometries.** **a** | Transmission electron micrograph of a dislocation point defect in an artificial square ice. The Burgers circuit is shown in blue and the Burgers vector *b* is shown in orange. **b** | Scanning electron microscopy (SEM) image of part of a chiral ice array, based on rotating all the nanomagnets in a square ice through an angle of 45°. **c** | SEM image of part of a shakti lattice. **d** | SEM image of part of a rhomboid, quasiperiodic Penrose tiling. Panel **a** is reprinted from REF.[37], CC-BY-4.0 https://creativecommons.org/licenses/by/4.0/. Panel **b** is reprinted from REF.[9], Springer Nature Limited. Panel **c** is reprinted from REF.[7], Springer Nature Limited. Panel **d**, image courtesy of Aaron Stein, Brookhaven National Laboratory, USA.

decagonal ordering in a Heisenberg system[145]. Alongside the lack of translational symmetry of the lattice, the quasi-1D nature of the skeleton enhances the frustration. The enhancement in frustration arises because 1D Ising systems do not order, and means that the ground state is very difficult to access, even using thermal annealing, offering a possible microscopic explanation for the glassy freezing in bulk magnetic quasicrystals.

As many of the examples above illustrate, topologically equivalent geometries can behave very differently because of different point symmetry at the vertices. Lower symmetry at the vertices leads to dissimilar interactions between the nanomagnets. For example, both the kagome and the brickwork lattice have vertices where three nanomagnets meet. However, the kagome lattice gives perfect degeneracy of the vertex states, whereas the macrospin configurations of the vertices in the brickwork lattice are non-degenerate because they are T-shaped. Thus, even though the pairwise interactions are frustrated in both systems, in the artificial kagome spin ice, any of the three interactions at the vertex can be frustrated, whereas in the brickwork lattice, it is favourable to arrange the frustration on parallel moments.

Outlook

We highlight here the key directions for future research in the field of artificial spin ice (FIG. 6). In terms of fundamental science, there are many more geometries and topologies to explore that can lead to the discovery of new, emergent phenomena. The exploration of the phase diagrams in artificial spin systems has only recently begun, spurred on by the creation of systems with coupled superparamagnets in thermodynamic equilibrium and the possibilities for characterization with low-energy muon spin spectroscopy[11], as well as neutron and synchrotron X-ray scattering. In particular, the use of indirect measurement techniques such as resonant soft X-ray scattering[12,97,151,152], neutron scattering and reflectometry[100–102,153], and X-ray photon correlation spectroscopy[82,154] can contribute to further understanding of the long-range and short-range spatial correlations in these frustrated systems. The low-temperature phases in artificial kagome spin ice have yet to be observed, and there are many different universality classes of phase transitions to explore, not only Ising but also Kosterlitz–Thouless[22,93,155] and Potts transitions[94], to name a few. For this, not only do the relevant spin-ice geometries need to be created but characterization methods need to be developed to measure magnetization dynamics at high spatial and temporal resolution, to quantify and classify the transitions unambiguously.

Improvements in nanofabrication methods will help to create new artificial spin-ice geometries, enabling the observation of novel, emergent phenomena. Indeed, the exploration of 3D structures, which bring additional degrees of freedom, has only just started[156,157]. One route to create 3D structures is to manufacture layered systems[41,43,158], applying the same lithography methods used to create 2D systems. Alternatively, 3D structures (BOX 2) can be manufactured with two-photon laser lithography[159–161] to create, for example, a buckyball structure[162] — a 2D connected spin-ice structure wrapped around on itself — and also to manufacture more complex structures[163]. Smaller 3D structures can be made with focused-electron-beam-induced deposition[164,165] and self-assembly methods[153,166,167]. Such technological advances mean that the designs of structures that exhibit new phenomena are only limited by imagination and ingenuity.

In terms of magnetic materials, to date, mainly ferromagnetic thin-film materials have been explored, but other materials offer additional functionality to artificial spin ice. For example, in nanomagnets fabricated from ferrimagnetic alloys[87], the compensation temperature can be tuned via the composition to give specific temperature-dependent behaviour of the magnetization. Alternatively, complex anisotropies can be introduced with single-crystal materials[24,168], which can also be adapted by modifying the nanomagnet shape[169]. Interesting behaviour can be obtained with hybrid thin-film systems combining, for example, different magnetic thin films, artificial spin ices with superconductors[170,171] or magnetostrictive and piezoelectric materials in so-called multiferroic composites[3].

There are many applications for which artificial spin ices have the potential to play an important role, because of the possibilities for non-volatile, low-power devices and the large number of macrospin configurations. These applications include computation, data storage and encryption, with prospects for their use in the emerging field of unconventional computing, including non-Boolean or neuromorphic computation[172]. For these applications, it will be important to find a way to access the large number of macrospin configurations[84]. Furthermore, to be compatible with CMOS (complementary metal–oxide–semiconductor) technology, it will be important to develop methods to write data electrically in the form of the magnetic state of individual or collections of magnets utilizing, for example, spin-orbit torques[173] or spin-transfer torques, and to read out the states of individual magnets or an array of magnets with transport measurements[66,99,174]. The magnetic state of individual nanomagnets could be read out electrically via magnetic tunnel junctions, for instance. In addition to electric currents, other stimuli that can be implemented to control the magnetic configurations include electric and magnetic fields[135], strain[58] and heat[57].

The first demonstrations of computation with arrays of magnets exploiting artificial spin-ice geometries have been performed[175–178], but there is still quite some way to go in terms of demonstrating functional devices that are CMOS-compatible. Other possibilities for devices include the exploitation of emergent magnetic monopoles, manipulating them in magnetic devices in the same way that the flow of electrons or spins is controlled in electric or spintronic devices. For this, basic control of monopole propagation has already been demonstrated by modifying the shape of individual nanomagnets (see supplementary information in REF.[6]). For neuromorphic computing, methods need to be found to mimic the spike-timing-dependent plasticity associated with neurons and, eventually, to create neural networks. For such bioinspired computation, it will be critical to implement an active or self-regulating control, for

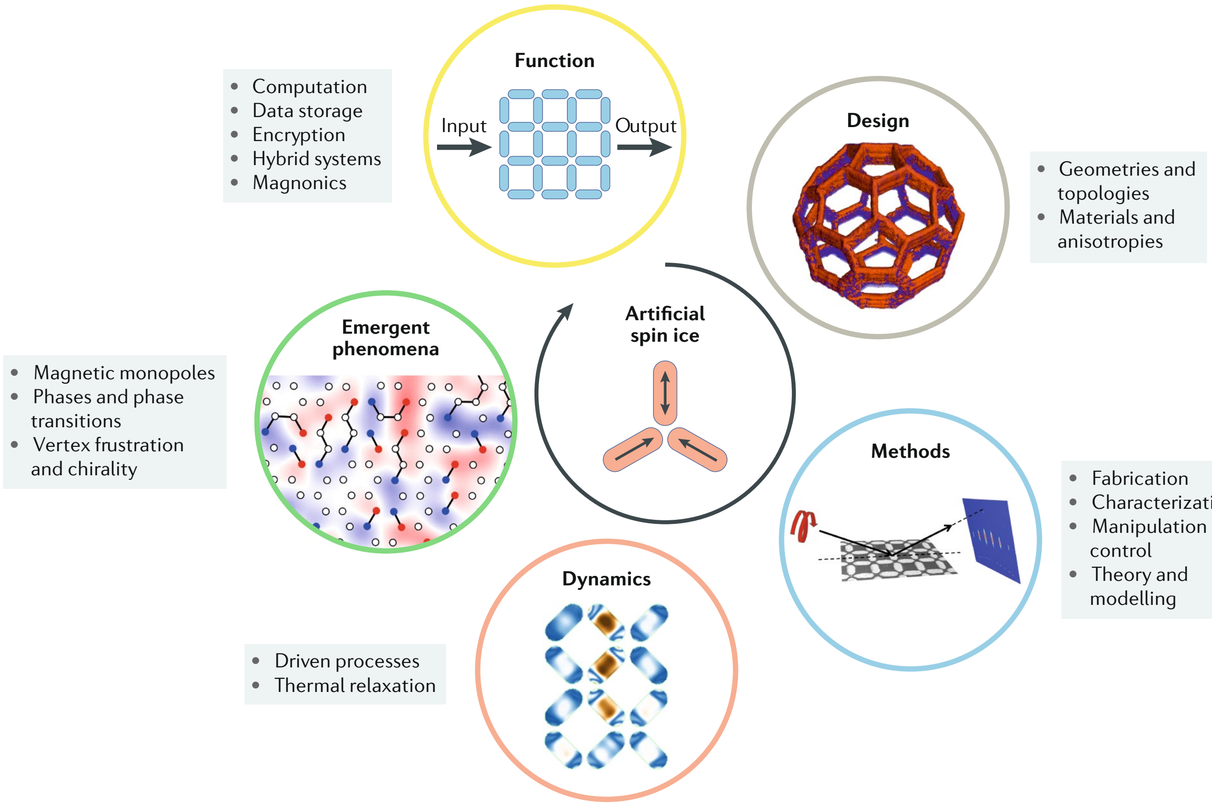


Fig. 6 | **The key directions for future research in artificial spin ice are interlinked.** Discoveries will be driven by inspiration for new designs. The new designs will require the development of methods to manufacture them and to measure their dynamics. In this way, it will be possible to identify, characterize and control the emergent phenomena that will lead to new functionality in artificial spin ice. 'Design' image is reprinted with permission from REF.[162], APS. 'Methods' image is adapted with permission from REF.[12], APS. 'Dynamics' image is adapted from REF.[52], APS. 'Emergent phenomena' image is adapted with permission from REF.[34], APS.

example by manipulating the magnetic state of individual nanomagnets through strain in a multiferroic composite[58]. It will also be important to be able to tune the interactions between the nanomagnets, something that is just starting to be implemented[48,49,173] and is likely to lead to new phenomena as well as new devices.

Mobile magnetic monopoles are also of interest for high-frequency dynamics, in the spirit of previous studies on pyrochlores, in which THz and GHz dynamics were used to investigate possibilities for monopole-generated responses[179]. The potential to control avalanche dynamics[6] provides a first step towards using artificial spin systems as platforms for reprogrammable magnonic resonators and crystals. The flexibility in design may allow, for example, the creation of artificial spin system geometries that can channel high-frequency excitations along a particular pathway as chosen by the trigger of a specific avalanche or some other mechanism to generate a desired magnetic configuration. Such devices would be of interest as elements in reconfigurable microwave circuits.

Moreover, concepts such as microwave-assisted reversal, which have been pursued for applications in data storage, can inspire new devices constructed from artificial spin systems. For example, GHz-frequency magnetic fields could be used to guide a frustrated artificial spin system into a specific magnetic configuration, which could be exploited in the logic and neuromorphic computing concepts mentioned above. Another route to manipulate magnetic configurations in artificial spin systems is to employ hybrid elements combining nanomagnets with thermoplasmonic heaters, allowing for fast, spatially selective and element-specific optical control of local temperatures[180].

For devices, both low-power operation and scalability are essential. For low-power applications, it might be feasible to create devices working at the Landauer limit[181]. In terms of scalability, it has been shown that artificial spin systems can be fabricated down to the atomic scale with the atoms precisely placed on a surface using a scanning tunnelling microscope tip[182]. While such atomic-scale artificial spin systems can only be implemented at low temperatures, the small length scales mean that the interactions can be modified, with the RKKY (Ruderman–Kittel–Kasuya–Yosida)-like interactions being ferromagnetic or antiferromagnetic, depending on the distance between the atoms. These atomic-scale systems may even provide a route to create analogies to the Kitaev model and quantum spin liquids.

Finally, it should be mentioned that artificial spin systems are not restricted to nanomagnets but can also be realized with colloids[183–187], superconductors[188,189] and buckled-polymer structures[190], and this list is by no means exhaustive. These non-magnetic systems have additional interesting phenomena of their own. For example, in colloidal ice, the ice rule can be destabilized by changing the topology of the square lattice with decimation. Here, effective magnetic charges can rearrange and give rise to charge screening[184]. Artificial spin-ice designs can also be used to create systems with a strong plasmonic response[191]. In addition, artificial spin ices made of nanomagnet arrays can be used to control other phenomena, including the motion of magnetic particles for biological applications[192], the behaviour of skyrmions[193,194] and the diffusion of vortices in superconducting materials[170].

The field of artificial spin ice has produced a number of interesting avenues for the future, which are guided and supported by theory and modelling. Inspiration for the field will continue to increase as the technology evolves to create artificial spin-ice geometries at ever-decreasing length scales and in 3D, to manufacture systems from magnetic materials with innovative properties, to manipulate them with different stimuli and to characterize the spatial correlations at ever-faster timescales.

Note added in proof At the time of publication, two important reviews on the dynamics in artificial spin ice[203] and on particle ice[204] have been published.

Published online: 08 November 2019

1. Wang, R. F. et al. Artificial 'spin ice' in a geometrically frustrated lattice of nanoscale ferromagnetic islands. *Nature* **439**, 303–306 (2006).
2. Harris, M. J., Bramwell, S. T., McMorrow, D. F., Zeiske, T. & Godfrey, K. W. Geometrical frustration in the ferromagnetic pyrochlore $Ho_2Ti_2O_7$. *Phys. Rev. Lett.* **79**, 2554–2557 (1997).
3. Heyderman, L. J. & Stamps, R. L. Artificial ferroic systems: novel functionality from structure, interactions and dynamics. *J. Phys. Condens. Matter* **25**, 363201 (2013).
4. Nisoli, C., Moessner, R. & Schiffer, P. Colloquium: Artificial spin ice: Designing and imaging magnetic frustration. *Rev. Mod. Phys.* **85**, 1473–1490 (2013).
5. Ladak, S., Read, D. E., Perkins, G. K., Cohen, L. F. & Branford, W. R. Direct observation of magnetic monopole defects in an artificial spin-ice system. *Nat. Phys.* **6**, 359–363 (2010).
6. Mengotti, E. et al. Real-space observation of emergent magnetic monopoles and associated Dirac strings in artificial kagome spin ice. *Nat. Phys.* **7**, 68–74 (2011).
7. Gilbert, I. et al. Emergent ice rule and magnetic charge screening from vertex frustration in artificial spin ice. *Nat. Phys.* **10**, 670–675 (2014).
8. Morrison, M. J., Nelson, T. R. & Nisoli, C. Unhappy vertices in artificial spin ice: new degeneracies from vertex frustration. *New J. Phys.* **15**, 045009 (2013).
9. Gliga, S. et al. Emergent dynamic chirality in a thermally driven artificial spin ratchet. *Nat. Mater.* **16**, 1106 (2017).
10. Kapaklis, V. et al. Melting artificial spin ice. *New J. Phys.* **14**, 035009 (2012).
11. Anghinolfi, L. et al. Thermodynamic phase transitions in a frustrated magnetic metamaterial. *Nat. Commun.* **6**, 8278 (2015).
12. Sendetskyi, O. et al. Continuous magnetic phase transition in artificial square ice. *Phys. Rev. B* **99**, 214430 (2019).
13. Krawczyk, M. & Grundler, D. Review and prospects of magnonic crystals and devices with reprogrammable band structure. *J. Phys. Condens. Matter* **26**, 123202 (2014).
14. Lao, Y. et al. Classical topological order in the kinetics of artificial spin ice. *Nat. Phys.* **14**, 723–727 (2018).
15. Gilbert, I. et al. Emergent reduced dimensionality by vertex frustration in artificial spin ice. *Nat. Phys.* **12**, 162–165 (2016).
16. Farhan, A. et al. Thermodynamics of emergent magnetic charge screening in artificial spin ice. *Nat. Commun.* **7**, 12635 (2016).
17. Farmer, B. et al. Direct imaging of coexisting ordered and frustrated sublattices in artificial ferromagnetic quasicrystals. *Phys. Rev. B* **93**, 134428 (2016).
18. Shi, D. et al. Frustration and thermalization in an artificial magnetic quasicrystal. *Nat. Phys.* **14**, 309–314 (2018).
19. Leo, N. et al. Collective magnetism in an artificial 2D XY spin system. *Nat. Commun.* **9**, 2850 (2018).
20. Schildknecht, D., Heyderman, L. J. & Derlet, P. M. Phase diagram of dipolar-coupled XY moments on disordered square lattices. *Phys. Rev. B* **98**, 064420 (2018).
21. Östman, E. et al. Interaction modifiers in artificial spin ices. *Nat. Phys.* **14**, 375–379 (2018).
22. Streubel, R. et al. Spatial and temporal correlations of XY macro spins. *Nano Lett.* **18**, 7428–7434 (2018).
23. Velten, S. et al. Vortex circulation patterns in planar microdisk arrays. *Appl. Phys. Lett.* **110**, 262406 (2017).
24. Louis, D. et al. A tunable magnetic metamaterial based on the dipolar four-state Potts model. *Nat. Mater.* **17**, 1076–1080 (2018).
25. Castelnovo, C., Moessner, R. & Sondhi, S. L. Magnetic monopoles in spin ice. *Nature* **451**, 42–45 (2008).
26. Fennell, T. et al. Magnetic Coulomb phase in the spin ice $Ho_2Ti_2O_7$. *Science* **326**, 415–417 (2009).
27. Kadowaki, H. et al. Observation of magnetic monopoles in spin ice. *J. Phys. Soc. Jpn.* **78**, 103706 (2009).
28. Morris, D. J. P. et al. Dirac strings and magnetic monopoles in the spin ice $Dy_2Ti_2O_7$. *Science* **326**, 411–414 (2009).
29. Bramwell, S. T. et al. Measurement of the charge and current of magnetic monopoles in spin ice. *Nature* **461**, 956–959 (2009).
30. Mól, L. A. et al. Magnetic monopole and string excitations in two-dimensional spin ice. *J. Appl. Phys.* **106**, 063913 (2009).
31. León, A. Heavy and light monopoles in magnetic reversion in artificial spin ice. *Curr. Appl. Phys.* **13**, 2014–2018 (2013).
32. Gilbert, I. et al. Direct visualization of memory effects in artificial spin ice. *Phys. Rev. B* **92**, 104417 (2015).
33. Farhan, A. et al. Direct observation of thermal relaxation in artificial spin ice. *Phys. Rev. Lett.* **111**, 057204 (2013).
34. Farhan, A., Derlet, P. M., Anghinolfi, L., Kleibert, A. & Heyderman, L. J. Magnetic charge and moment dynamics in artificial kagome spin ice. *Phys. Rev. B* **96**, 064409 (2017).
35. Silva, R. C. et al. Nambu monopoles interacting with lattice defects in a two-dimensional artificial square spin ice. *Phys. Rev. B* **87**, 014414 (2013).
36. Thonig, D. & Henk, J. Pinning of thermal excitations at defects in artificial dipolar arrays: A theoretical investigation. *J. Magn. Magn. Mater.* **386**, 117–124 (2015).
37. Drisko, J., Marsh, T. & Cumings, J. Topological frustration of artificial spin ice. *Nat. Commun.* **8**, 14009 (2017).
38. Vedmedenko, E. Y. Dynamics of bound monopoles in artificial spin ice: How to store energy in Dirac strings. *Phys. Rev. Lett.* **116**, 077202 (2016).
39. Budrikis, Z. et al. Domain dynamics and fluctuations in artificial square ice at finite temperatures. *New J. Phys.* **14**, 035014 (2012).
40. Mól, L. A. S., Moura-Melo, W. A. & Pereira, A. R. Conditions for free magnetic monopoles in nanoscale square arrays of dipolar spin ice. *Phys. Rev. B* **82**, 054434 (2010).
41. Chern, G.-W., Reichhardt, C. & Nisoli, C. Realizing three-dimensional artificial spin ice by stacking planar nano-arrays. *Appl. Phys. Lett.* **104**, 013101 (2014).
42. Möller, G. & Moessner, R. Artificial square ice and related dipolar nanoarrays. *Phys. Rev. Lett.* **96**, 237202 (2006).
43. Perrin, Y., Canals, B. & Rougemaille, N. Extensive degeneracy, Coulomb phase and magnetic monopoles in artificial square ice. *Nature* **540**, 410–413 (2016).
44. Farhan, A. et al. Emergent magnetic monopole dynamics in macroscopically degenerate artificial spin ice. *Sci. Adv.* **5**, eaav6380 (2019).
45. Henley, C. L. The "Coulomb phase" in frustrated systems. *Annu. Rev. Condens. Matter Phys.* **1**, 179–210 (2010).
46. Ribeiro, I. R. B. et al. Realization of rectangular artificial spin ice and direct observation of high energy topology. *Sci. Rep.* **7**, 13982 (2017).
47. Loreto, R. P. et al. Experimental and theoretical evidences for the ice regime in planar artificial spin ices. *J. Phys. Condens. Matter* **31**, 025301 (2019).
48. Perrin, Y., Canals, B. & Rougemaille, N. Quasidegenerate ice manifold in a purely two-dimensional square array of nanomagnets. *Phys. Rev. B* **99**, 224434 (2019).
49. Schanilec, V., Perrin, Y., Denmat, S. L., Canals, B. & Rougemaille, N. Artificial vertex systems by design. Preprint at *arXiv* https://arxiv.org/abs/1902.00452 (2019).
50. Takatsu, H. et al. Two-dimensional monopole dynamics in the dipolar spin ice $Dy_2Ti_2O_7$. *J. Phys. Soc. Jpn.* **82**, 073707 (2013).
51. Otsuka, H., Takatsu, H., Goto, K. & Kadowaki, H. Scaling ansatz for the ac magnetic response in two-dimensional spin ice. *Phys. Rev. B* **90**, 144428 (2014).
52. Gliga, S., Kákay, A., Hertel, R. & Heinonen, O. G. Spectral analysis of topological defects in an artificial spin-ice lattice. *Phys. Rev. Lett.* **110**, 117205 (2013).
53. Bhat, V. S., Heimbach, F., Stasinopoulos, I. & Grundler, D. Magnetization dynamics of topological defects and the spin solid in a kagome artificial spin ice. *Phys. Rev. B* **93**, 140401 (2016).
54. Bhat, V. S., Heimbach, F., Stasinopoulos, I. & Grundler, D. Angular-dependent magnetization dynamics of kagome artificial spin ice incorporating topological defects. *Phys. Rev. B* **96**, 014426 (2017).
55. Chern, G.-W. & Mellado, P. Magnetic monopole polarons in artificial spin ices. *EPL* **114**, 37004 (2016).
56. Loreto, R. P. et al. Emergence and mobility of monopoles in a unidirectional arrangement of magnetic nanoislands. *Nanotechnology* **26**, 295303 (2015).
57. León, A. Thermal phase transition in artificial spin ice systems induces the formation and migration of monopole-like magnetic excitations. *Physica B Condens. Matter* **500**, 59–65 (2016).
58. Chavez, A. C., Barra, A. & Carman, G. P. Voltage control of magnetic monopoles in artificial spin ice. *J. Phys. D Appl. Phys.* **51**, 234001 (2018).
59. Mellado, P., Petrova, O., Shen, Y. & Tchernyshyov, O. Dynamics of magnetic charges in artificial spin ice. *Phys. Rev. Lett.* **105**, 187206 (2010).
60. Zeissler, K. et al. The non-random walk of chiral magnetic charge carriers in artificial spin ice. *Sci. Rep.* **3**, 1252 (2013).
61. Pushp, A. et al. Domain wall trajectory determined by its fractional topological edge defects. *Nat. Phys.* **9**, 505–511 (2013).
62. Omari, K. A. & Hayward, T. J. Chirality-based vortex domain-wall logic gates. *Phys. Rev. Applied* **2**, 044001 (2014).
63. Garg, C. et al. Highly asymmetric chiral domain-wall velocities in Y-shaped junctions. *Nano Lett.* **18**, 1826–1830 (2018).
64. Burn, D. M., Chadha, M., Walton, S. K. & Branford, W. R. Dynamic interaction between domain walls and nanowire vertices. *Phys. Rev. B* **90**, 144414 (2014).
65. Walton, S. K. et al. Limitations in artificial spin ice path selectivity: the challenges beyond topological control. *New J. Phys.* **17**, 013054 (2015).
66. Burn, D. M., Chadha, M. & Branford, W. R. Dynamic dependence to domain wall propagation through artificial spin ice. *Phys. Rev. B* **95**, 104417 (2017).
67. Burn, D. M., Chadha, M. & Branford, W. R. Angular-dependent magnetization reversal processes in artificial spin ice. *Phys. Rev. B* **92**, 214425 (2015).

68. Kwon, J. et al. Low field domain wall dynamics in artificial spin-ice basis structure. *J. Appl. Phys.* **118**, 163907 (2015).
69. Sethi, P. et al. Direct observation of deterministic domain wall trajectory in magnetic network structures. *Sci. Rep.* **6**, 19027 (2016).
70. Gartside, J. C. et al. Realization of ground state in artificial kagome spin ice via topological defect-driven magnetic writing. *Nat. Nanotechnol.* **13**, 53–58 (2018).
71. Rougemaille, N. et al. Chiral nature of magnetic monopoles in artificial spin ice. *New J. Phys.* **15**, 035026 (2013).
72. Nisoli, C. et al. Effective temperature in an interacting vertex system: theory and experiment on artificial spin ice. *Phys. Rev. Lett.* **105**, 047205 (2010).
73. Morgan, J. P. et al. Real and effective thermal equilibrium in artificial square spin ices. *Phys. Rev. B* **87**, 024405 (2013).
74. Lammert, P. E. et al. Direct entropy determination and application to artificial spin ice. *Nat. Phys.* **6**, 786–789 (2010).
75. Porro, J. M., Bedoya-Pinto, A., Berger, A. & Vavassori, P. Exploring thermally induced states in square artificial spin-ice arrays. *New J. Phys.* **15**, 055012 (2013).
76. Zhang, S. et al. Crystallites of magnetic charges in artificial spin ice. *Nature* **500**, 553–557 (2013).
77. Drisko, J., Daunheimer, S. & Cumings, J. $FePd_3$ as a material for studying thermally active artificial spin ice systems. *Phys. Rev. B* **91**, 224406 (2015).
78. Ke, X. et al. Energy minimization and ac demagnetization in a nanomagnet array. *Phys. Rev. Lett.* **101**, 037205 (2008).
79. Budrikis, Z. et al. Disorder strength and field-driven ground state domain formation in artificial spin ice: experiment, simulation, and theory. *Phys. Rev. Lett.* **109**, 037203 (2012).
80. Coey, J. M. D. *Magnetism and Magnetic Materials* (Cambridge Univ. Press, 2010).
81. Osborn, J. A. Demagnetizing factors of the general ellipsoid. *Phys. Rev.* **67**, 351–357 (1945).
82. Morley, S. A. et al. Vogel-Fulcher-Tammann freezing of a thermally fluctuating artificial spin ice probed by x-ray photon correlation spectroscopy. *Phys. Rev. B* **95**, 104422 (2017).
83. Andersson, M. S. et al. Thermally induced magnetic relaxation in square artificial spin ice. *Sci. Rep.* **6**, 37097 (2016).
84. Farhan, A. et al. Exploring hyper-cubic energy landscapes in thermally active finite artificial spin-ice systems. *Nat. Phys.* **9**, 375–382 (2013).
85. Kapaklis, V. et al. Thermal fluctuations in artificial spin ice. *Nat. Nanotechnol.* **9**, 514–519 (2014).
86. Morley, S. A. et al. Effect of FePd alloy composition on the dynamics of artificial spin ice. *Sci. Rep.* **8**, 4750 (2018).
87. Montaigne, F. et al. Size distribution of magnetic charge domains in thermally activated but out-of-equilibrium artificial spin ice. *Sci. Rep.* **4**, 5702 (2014).
88. Rougemaille, N. & Canals, B. Cooperative magnetic phenomena in artificial spin systems: spin liquids, Coulomb phase and fragmentation of magnetism – a colloquium. *Eur. Phys. J. B* **92**, 62 (2019).
89. Morgan, J. P., Stein, A., Langridge, S. & Marrows, C. H. Thermal ground-state ordering and elementary excitations in artificial magnetic square ice. *Nat. Phys.* **7**, 75–79 (2011).
90. Silva, R. C., Nascimento, F. S., Mól, L. A. S., Moura-Melo, W. A. & Pereira, A. R. Thermodynamics of elementary excitations in artificial magnetic square ice. *New J. Phys.* **14**, 015008 (2012).
91. Levis, D., Cugliandolo, L. F., Foini, L. & Tarzia, M. Thermal phase transitions in artificial spin ice. *Phys. Rev. Lett.* **110**, 207206 (2013).
92. Parakkat, V. M., Xie, K. & Krishnan, K. M. Tunable ground state in heterostructured artificial spin ice with exchange bias. *Phys. Rev. B* **99**, 054429 (2019).
93. Chern, G.-W., Mellado, P. & Tchernyshyov, O. Two-stage ordering of spins in dipolar spin ice on the kagome lattice. *Phys. Rev. Lett.* **106**, 207202 (2011).
94. Chern, G.-W. & Tchernyshyov, O. Magnetic charge and ordering in kagome spin ice. *Phil. Trans. R. Soc. A* **370**, 5718–5737 (2012).
95. Canals, B. et al. Fragmentation of magnetism in artificial kagome dipolar spin ice. *Nat. Commun.* **7**, 11446 (2016).
96. Brooks-Bartlett, M. E., Banks, S. T., Jaubert, L. D. C., Harman-Clarke, A. & Holdsworth, P. C. W. Magnetic-moment fragmentation and monopole crystallization. *Phys. Rev. X* **4**, 011007 (2014).
97. Sendetskyi, O. et al. Magnetic diffuse scattering in artificial kagome spin ice. *Phys. Rev. B* **93**, 224413 (2016).
98. Farhan, A. et al. Thermally induced magnetic relaxation in building blocks of artificial kagome spin ice. *Phys. Rev. B* **89**, 214405 (2014).
99. Branford, W. R., Ladak, S., Read, D. E., Zeissler, K. & Cohen, L. F. Emerging chirality in artificial spin ice. *Science* **335**, 1597–1600 (2012).
100. Summers, B., Chen, Y., Dahal, A. & Singh, D. K. New description of evolution of magnetic phases in artificial honeycomb lattice. *Sci. Rep.* **7**, 16080 (2017).
101. Glavic, A. et al. Spin solid versus magnetic charge ordered state in artificial honeycomb lattice of connected elements. *Adv. Sci*. **5**, 1700856 (2018).
102. Summers, B. et al. Temperature-dependent magnetism in artificial honeycomb lattice of connected elements. *Phys. Rev. B* **97**, 014401 (2018).
103. Chioar, I. A. et al. Kinetic pathways to the magnetic charge crystal in artificial dipolar spin ice. *Phys. Rev. B* **90**, 220407 (2014).
104. Liashko, S. Y., Jónsson, H. & Uzdin, V. M. The effect of temperature and external field on transitions in elements of kagome spin ice. *New J. Phys.* **19**, 113008 (2017).
105. Petersen, C. F. et al. Tuning magnetic ordering in a dipolar square-kite tessellation. *Appl. Phys. Lett.* **112**, 092403 (2018).
106. Chioar, I. A. et al. Nonuniversality of artificial frustrated spin systems. *Phys. Rev. B* **90**, 064411 (2014).
107. Hamp, J., Moessner, R. & Castelnovo, C. Supercooling and fragile glassiness in a dipolar kagome Ising magnet. *Phys. Rev. B* **98**, 144439 (2018).
108. Sklenar, J. et al. Field-induced phase coexistence in an artificial spin ice. *Nat. Phys.* **15**, 191–195 (2018).
109. Carlotti, G. et al. From micro- to nanomagnetic dots: evolution of the eigenmode spectrum on reducing the lateral size. *J. Phys. D Appl. Phys.* **47**, 265001 (2014).
110. Li, Y. et al. Thickness dependence of spin wave excitations in an artificial square spin ice-like geometry. *J. Appl. Phys.* **121**, 103903 (2017).
111. Sklenar, J., Bhat, V. S., DeLong, L. E. & Ketterson, J. B. Broadband ferromagnetic resonance studies on an artificial square spin-ice island array. *J. Appl. Phys.* **113**, 17B530 (2013).
112. Ma, F., Wu, Y. & Zong, B. Micromagnetic investigation of microwave permeability of magnetic artificial spin ice. *Mater. Sci. Appl.* **5**, 991–995 (2014).
113. Montoncello, F. et al. Mutual influence between macrospin reversal order and spin-wave dynamics in isolated artificial spin-ice vertices. *Phys. Rev. B* **97**, 014421 (2018).
114. Mironov, V. L., Skorohodov, E. V. & Blackman, J. A. Magnetic states and ferromagnetic resonance in geometrically frustrated arrays of multilayer ferromagnetic nanoparticles ordered on triangular lattices. *J. Appl. Phys.* **115**, 184301 (2014).
115. Gliga, S., Kákay, A., Heyderman, L. J., Hertel, R. & Heinonen, O. G. Broken vertex symmetry and finite zero-point entropy in the artificial square ice ground state. *Phys. Rev. B* **92**, 060413 (2015).
116. Behncke, C. et al. Tunable geometrical frustration in magnonic vortex crystals. *Sci. Rep*. **8**, 186 (2018).
117. Jungfleisch, M. B. et al. High-frequency dynamics modulated by collective magnetization reversal in artificial spin ice. *Phys. Rev. Applied* **8**, 064026 (2017).
118. Bhat, V. S. & Grundler, D. Angle-dependent magnetization dynamics with mirror-symmetric excitations in artificial quasicrystalline nanomagnet lattices. *Phys. Rev. B* **98**, 174408 (2018).
119. Kruglyak, V. V. et al. Formation of the band spectrum of spin waves in 1D magnonic crystals with different types of interfacial boundary conditions. *J. Phys. D Appl. Phys.* **50**, 094003 (2017).
120. Szulc, K. et al. Remagnetization in arrays of ferromagnetic nanostripes with periodic and quasiperiodic order. *Phys. Rev. B* **99**, 064412 (2019).
121. Lisiecki, F. et al. Reprogrammability and scalability of magnonic Fibonacci quasicrystals. *Phys. Rev. Applied* **11**, 054003 (2019).
122. Iacocca, E., Gliga, S., Stamps, R. L. & Heinonen, O. Reconfigurable wave band structure of an artificial square ice. *Phys. Rev. B* **93**, 134420 (2016).
123. Iacocca, E. & Heinonen, O. Topologically nontrivial magnon bands in artificial square spin ices with Dzyaloshinskii-Moriya interaction. *Phys. Rev. Applied* **8**, 034015 (2017).
124. Zhou, X., Chua, G. L., Singh, N. & Adeyeye Adekunle, O. Large area artificial spin ice and anti-spin ice $Ni_{80}Fe_{20}$ structures: static and dynamic behavior. *Adv. Funct. Mater.* **26**, 1437–1444 (2016).
125. Schneider, T. et al. Programmability of Co-antidot lattices of optimized geometry. *Sci. Rep.* **7**, 41157 (2017).
126. Mamica, S., Krawczyk, M. & Grundler, D. Nonuniform spin-wave softening in two-dimensional magnonic crystals as a tool for opening omnidirectional magnonic band gaps. *Phys. Rev. Applied* **11**, 054011 (2019).
127. Mamica, S., Zhou, X., Adeyeye, A., Krawczyk, M. & Gubbiotti, G. Spin-wave dynamics in artificial anti-spin-ice systems: Experimental and theoretical investigations. *Phys. Rev. B* **98**, 054405 (2018).
128. Grundler, D. Nanomagnonics. *J. Phys. D Appl. Phys.* **49**, 391002 (2016).
129. Li, J. et al. Comparing artificial frustrated magnets by tuning the symmetry of nanoscale permalloy arrays. *Phys. Rev. B* **81**, 092406 (2010).
130. Farhan, A. et al. Nanoscale control of competing interactions and geometrical frustration in a dipolar trident lattice. *Nat. Commun.* **8**, 995 (2017).
131. Macêdo, R., Macauley, G. M., Nascimento, F. S. & Stamps, R. L. Apparent ferromagnetism in the pinwheel artificial spin ice. *Phys. Rev. B* **98**, 014437 (2018).
132. Li, Y. et al. Superferromagnetism and domain-wall topologies in artificial "pinwheel" spin ice. *ACS Nano* **13**, 2213–2222 (2019).
133. Yu, L. et al. Monte Carlo simulation on a new artificial spin ice lattice consisting of hexagons and three-moment vertices. *AIP Adv.* **7**, 085211 (2017).
134. Lehmann, J., Donnelly, C., Derlet, P. M., Heyderman, L. J. & Fiebig, M. Poling of an artificial magneto-toroidal crystal. *Nat. Nanotechnol.* **14**, 141–144 (2018).
135. Wang, Y.-L. et al. Rewritable artificial magnetic charge ice. *Science* **352**, 962–966 (2016).
136. Chern, G.-W., Morrison, M. J. & Nisoli, C. Degeneracy and criticality from emergent frustration in artificial spin ice. *Phys. Rev. Lett.* **111**, 177201 (2013).
137. Stopfel, H. et al. Magnetic order and energy-scale hierarchy in artificial spin-ice structures. *Phys. Rev. B* **98**, 014435 (2018).
138. Nisoli, C. in *Frustrated Materials and Ferroic Glasses* (eds Lookman, T. & Ren, X.) 57-99 (Springer International Publishing, 2018).
139. Stamps, R. L. Artificial spin ice: The unhappy wanderer. *Nat. Phys.* **10**, 623–624 (2014).
140. Mydosh, J. A. *Spin Glasses: An Experimental Introduction*. (Taylor & Francis, 1993).
141. Saccone, M. et al. Towards artificial Ising spin glasses: thermal ordering in randomized arrays of Ising-type nanomagnets. *Phys. Rev. B* **99**, 224403 (2019).
142. Islam, Z. et al. Reinvestigation of long-range magnetic ordering in icosahedral Tb-Mg-Zn. *Phys. Rev. B* **57**, R11047–R11050 (1998).
143. Chernikov, M. A., Bernasconi, A., Beeli, C., Schilling, A. & Ott, H. R. Low-temperature magnetism in icosahedral $Al_{70}Mn_9Pd_{21}$. *Phys. Rev. B* **48**, 3058–3065 (1993).
144. Penrose, R. The role of aesthetics in pure and applied mathematical research. *Bull. Inst. Math. Appl*. **10**, 266-271 (1974).
145. Vedmedenko, E. Y., Oepen, H. P. & Kirschner, J. Decagonal quasiferromagnetic microstructure on the Penrose tiling. *Phys. Rev. Lett.* **90**, 137203 (2003).
146. Vedmedenko, E. Y., Grimm, U. & Wiesendanger, R. Interplay between magnetic and spatial order in quasicrystals. *Philosoph. Mag.* **86**, 733–739 (2006).
147. Bhat, V. S. et al. Controlled magnetic reversal in permalloy films patterned into artificial quasicrystals. *Phys. Rev. Lett.* **111**, 077201 (2013).
148. Brajuskovic, V., Barrows, F., Phatak, C. & Petford-Long, A. K. Real-space observation of magnetic excitations and avalanche behavior in artificial quasicrystal lattices. *Sci. Rep.* **6**, 34384 (2016).
149. Brajuskovic, V., Addi, A., Phatak, C. & Petford-Long, A. K. Observation of transient states during magnetization reversal in a quasicrystal artificial spin ice. *Phys. Rev. B* **98**, 094424 (2018).
150. Barrows, F., Brajuskovic, V., Petford-Long, A. K. & Phatak, C. Emergent magnetic ordering and topological frustration in quasicrystal artificial spin ices. *Phys. Rev. B* **99**, 094424 (2019).
151. Perron, J. et al. Extended reciprocal space observation of artificial spin ice with x-ray resonant magnetic scattering. *Phys. Rev. B* **88**, 214424 (2013).
152. Lee, J. C. T. et al. Textured heterogeneity in square artificial spin ice. *Phys. Rev. B* **99**, 024406 (2019).
153. Mistonov, A. A. et al. Magnetic structure of the inverse opal-like structures: small angle neutron diffraction

and micromagnetic simulations. *J. Magn. Magn. Mater.* **477**, 99–108 (2019).
154. Chen, X. M. et al. Spontaneous magnetic superdomain wall fluctuations in an artificial antiferromagnet. Preprint at *arXiv* https://arxiv.org/abs/1809.05656 (2018).
155. Santos-Filho, J. B., Plascak, J. A., Sobrinho, M. C. & Araujo Batista, T. S. Phase diagram of the XY Vector Blume–Emery–Griffiths model on a Kagome lattice by Monte Carlo simulation. *Physica A Stat. Mech. Appl.* **503**, 844–848 (2018).
156. Streubel, R. et al. Magnetism in curved geometries. *J. Phys. D Appl. Phys.* **49**, 363001 (2016).
157. Fernández-Pacheco, A. et al. Three-dimensional nanomagnetism. *Nat. Commun.* **8**, 15756 (2017).
158. Lavrijsen, R. et al. Magnetic ratchet for three-dimensional spintronic memory and logic. *Nature* **493**, 647–650 (2013).
159. Kadic, M., Bückmann, T., Stenger, N., Thiel, M. & Wegener, M. On the practicability of pentamode mechanical metamaterials. *Appl. Phys. Lett.* **100**, 191901 (2012).
160. Williams, G. et al. Two-photon lithography for 3D magnetic nanostructure fabrication. *Nano Res.* **11**, 845–854 (2018).
161. May, A., Hunt, M., Berg, A. V. D., Hejazi, A. & Ladak, S. Realisation of a frustrated 3D magnetic nanowire lattice. *Commun. Phys.* **2**, 13 (2019).
162. Donnelly, C. et al. Element-specific X-ray phase tomography of 3D structures at the nanoscale. *Phys. Rev. Lett.* **114**, 115501 (2015).
163. Frenzel, T., Kadic, M. & Wegener, M. Three-dimensional mechanical metamaterials with a twist. *Science* **358**, 1072–1074 (2017).
164. Keller, L. et al. Direct-write of free-form building blocks for artificial magnetic 3D lattices. *Sci. Rep.* **8**, 6160 (2018).
165. Fowlkes, J. D. et al. High-fidelity 3D-nanoprinting via focused electron beams: computer-aided design (3BID). *ACS Appl. Nano Mater.* **1**, 1028–1041 (2018).
166. Mistonov, A. A. et al. Three-dimensional artificial spin ice in nanostructured Co on an inverse opal-like lattice. *Phys. Rev. B* **87**, 220408 (2013).
167. Shishkin, I. S. et al. Nonlinear geometric scaling of coercivity in a three-dimensional nanoscale analog of spin ice. *Phys. Rev. B* **94**, 064424 (2016).
168. Chopdekar, R. V. et al. Nanostructured complex oxides as a route towards thermal behavior in artificial spin ice systems. *Phys. Rev. Mater.* **1**, 024401 (2017).
169. Cowburn, R. P. Property variation with shape in magnetic nanoelements. *J. Phys. D Appl. Phys.* **33**, R1 (2000).
170. Wang, Y.-L. et al. Switchable geometric frustration in an artificial-spin-ice–superconductor heterosystem. *Nat. Nanotechnol.* **13**, 560–565 (2018).
171. Rollano, V. et al. Topologically protected superconducting ratchet effect generated by spin-ice nanomagnets. *Nanotechnology* **30**, 244003 (2019).
172. Torrejon, J. et al. Neuromorphic computing with nanoscale spintronic oscillators. *Nature* **547**, 428–431 (2017).
173. Luo, Z. et al. Chirally coupled nanomagnets. *Science* **363**, 1435–1439 (2019).
174. Le, B. L. et al. Understanding magnetotransport signatures in networks of connected permalloy nanowires. *Phys. Rev. B* **95**, 060405 (2017).
175. Gypens, P., Leliaert, J. & Van Waeyenberge, B. Balanced magnetic logic gates in a kagome spin ice. *Phys. Rev. Applied* **9**, 034004 (2018).
176. Caravelli, F. & Nisoli, C. Computation via interacting magnetic memory bites: integration of boolean gates. Preprint at *arXiv* https://arxiv.org/abs/1810.09190 (2018).
177. Arava, H. et al. Engineering relaxation pathways in building blocks of artificial spin ice for computation. *Phys. Rev. Applied* **11**, 054086 (2019).
178. Arava, H. et al. Computational logic with square rings of nanomagnets. *Nanotechnology* **29**, 265205 (2018).
179. Pan, L., Laurita, N. J., Ross, K. A., Gaulin, B. D. & Armitage, N. P. A measure of monopole inertia in the quantum spin ice $Yb_2Ti_2O_7$. *Nat. Phys.* **12**, 361–366 (2016).
180. Pancaldi, M., Leo, N. & Vavassori, P. Selective and fast plasmon-assisted photo-heating of nanomagnets. *Nanoscale* **11**, 7656–7666 (2019).
181. Hong, J., Lambson, B., Dhuey, S. & Bokor, J. Experimental test of Landauer's principle in single-bit operations on nanomagnetic memory bits. *Sci. Adv.* **2**, e1501492 (2016).
182. Khajetoorians, A. A. et al. Atom-by-atom engineering and magnetometry of tailored nanomagnets. *Nat. Phys.* **8**, 497–503 (2012).
183. Libál, A., Reichhardt, C. & Reichhardt, C. J. O. Realizing colloidal artificial ice on arrays of optical traps. *Phys. Rev. Lett.* **97**, 228302 (2006).
184. Libál, A. et al. Ice rule fragility via topological charge transfer in artificial colloidal ice. *Nat. Commun.* **9**, 4146 (2018).
185. Han, Y. et al. Geometric frustration in buckled colloidal monolayers. *Nature* **456**, 898–903 (2008).
186. Le Cunuder, A., Frérot, I., Ortiz-Ambriz, A. & Tierno, P. Competing orders in colloidal kagome ice: Importance of the in-trap motion of the particles. *Phys. Rev. B* **99**, 140405 (2019).
187. Nisoli, C. Unexpected phenomenology in particle-based ice absent in magnetic spin ice. *Phys. Rev. Lett.* **120**, 167205 (2018).
188. Libál, A., Reichhardt, C. J. O. & Reichhardt, C. Creating artificial ice states using vortices in nanostructured superconductors. *Phys. Rev. Lett.* **102**, 237004 (2009).
189. Xue, C. et al. Tunable artificial vortex ice in nanostructured superconductors with a frustrated kagome lattice of paired antidots. *Phys. Rev. B* **97**, 134506 (2018).
190. Kang, S. H. et al. Complex ordered patterns in mechanical instability induced geometrically frustrated triangular cellular structures. *Phys. Rev. Lett.* **112**, 098701 (2014).
191. Conde-Rubio, A. et al. Geometric frustration in a hexagonal lattice of plasmonic nanoelements. *Opt. Express* **26**, 20211–20224 (2018).
192. Gunnarsson, K. et al. Programmable motion and separation of single magnetic particles on patterned magnetic surfaces. *Adv. Mater.* **17**, 1730–1734 (2005).
193. Fert, A., Cros, V. & Sampaio, J. Skyrmions on the track. *Nat. Nanotechnol.* **8**, 152–156 (2013).
194. Ma, F., Reichhardt, C., Gan, W., Reichhardt, C. J. O. & Lew, W. S. Emergent geometric frustration of artificial magnetic skyrmion crystals. *Phys. Rev. B* **94**, 144405 (2016).
195. Braun, H.-B. Topological effects in nanomagnetism: from superparamagnetism to chiral quantum solitons. *Adv. Phys.* **61**, 1–116 (2012).
196. Ryzhkin, I. A. Magnetic relaxation in rare-earth oxide pyrochlores. *J. Exp. Theor. Phys.* **101**, 481–486 (2005).
197. Park, J. et al. Magnetic response of brickwork artificial spin ice. *Phys. Rev. B* **96**, 024436 (2017).
198. Pohle, R., Benton, O. & Jaubert, L. D. C. Reentrance of disorder in the anisotropic shuriken Ising model. *Phys. Rev. B* **94**, 014429 (2016).
199. Arnalds, U. B. et al. A new look on the two-dimensional Ising model: thermal artificial spins. *New J. Phys.* **18**, 023008 (2016).
200. Liu, Y. et al. Confined chemical fluid deposition of ferromagnetic metalattices. *Nano Lett.* **18**, 546–552 (2018).
201. Chioar, I. A., Rougemaille, N. & Canals, B. Ground-state candidate for the classical dipolar kagome Ising antiferromagnet. *Phys. Rev. B* **93**, 214410 (2016).
202. Zhang, S. et al. Perpendicular magnetization and generic realization of the Ising model in artificial spin ice. *Phys. Rev. Lett.* **109**, 087201 (2012).
203. Lendinez, S. & Jungfleisch, M. B. Magnetization dynamics in artificial spin ice. *J. Phys. Condens. Matter* **32**, 013001 (2020).
204. Ortiz-Ambriz, A. Nisoli, C., Reichhardt, C., Reichhardt, C. J. O. & Tierno, P. Ice rule and emergent frustration in particle ice and beyond. Preprint at *arXiv* https://arxiv.org/abs/1909.13534 (2019).

Acknowledgements
The authors are very grateful for the important insights from P. M. Derlet, H.-B. Braun, D. Grundler, V. Scagnoli, K. Hofhuis and Q. N. Meier, which were of great value in the preparation of this manuscript. L.J.H. and S.H.S. acknowledge financial support provided by the Swiss National Science Foundation, grant no. 200020_172774.

Author contributions
All authors have read, discussed and contributed to the writing of the manuscript.

Competing interests
The authors declare no competing interests.

Publisher's note
Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Reviewer information
Nature Reviews Physics thanks C. Nisoli, B. Hjörvarsson and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.